Introduction to Quantile Regression

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Motivation

What the regression curve does is give a grand summary for the averages of the distributions corresponding to the set of of x's. We could go further and compute several different regression curves corresponding to the various percentage points of the distributions and thus get a more complete picture of the set.

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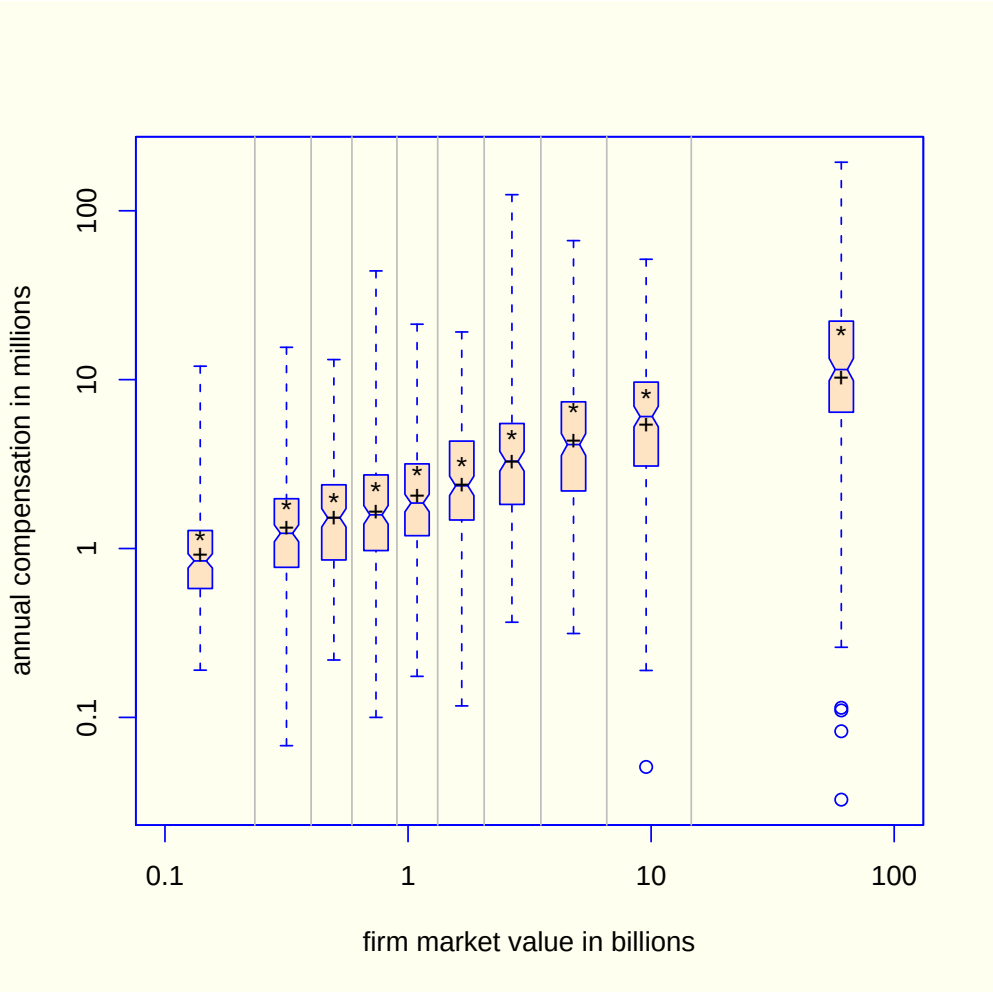
Mosteller and Tukey (1977)

Boxplot of CEO Pay by Firm Size



An Outline

- An Historical Introduction to Regression
- What is Quantile Regression?
- Beyond Average Treatment Effects
- Two Artificial Examples
- Three Introductory Empirical Examples
  - The Classical Engel Curve
  - A Model of Infant Birthweight
  - Maximum Daily Temperature in Melbourne

Boscovich's Problem

Boscovich (1757) proposed estimating the ellipticity of the earth by solving:

$$\min \sum_{i=1}^{n} |y_i - \alpha - x_i'\beta|$$

subject to:

$$\sum_{i=1}^{n} (y_i - \alpha - x_i'\beta) = 0,$$

The constraint allowed the problem to be reduced to the weighted median problem:

$$\min \sum_{i=1}^{n} |\tilde{y}_i - \tilde{x}_i'\beta|$$

where $\tilde{y}$ and $\tilde{x}$ denote deviations from their means.

Laplace's *Methode de Situation*

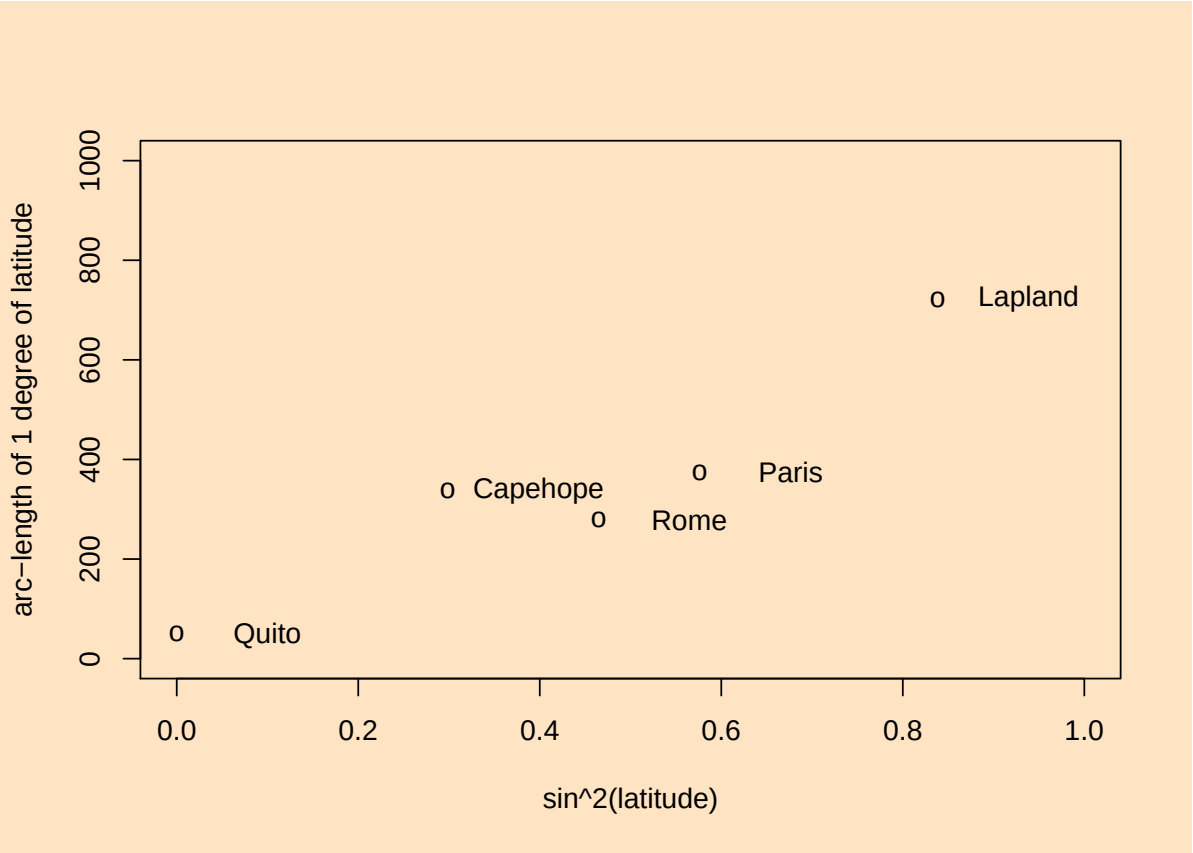
Laplace showed that Boscovich's problem:

$$\min \sum_{i=1}^{n} |\tilde{y}_i - \tilde{x}_i' \beta|$$

could be solved by ordering the candidate slopes $\{\tilde{y}_i / \tilde{x}_i\}$ and finding the weighted median using weights $w_i = |\tilde{x}_i|$, i.e. finding the smallest j such that,

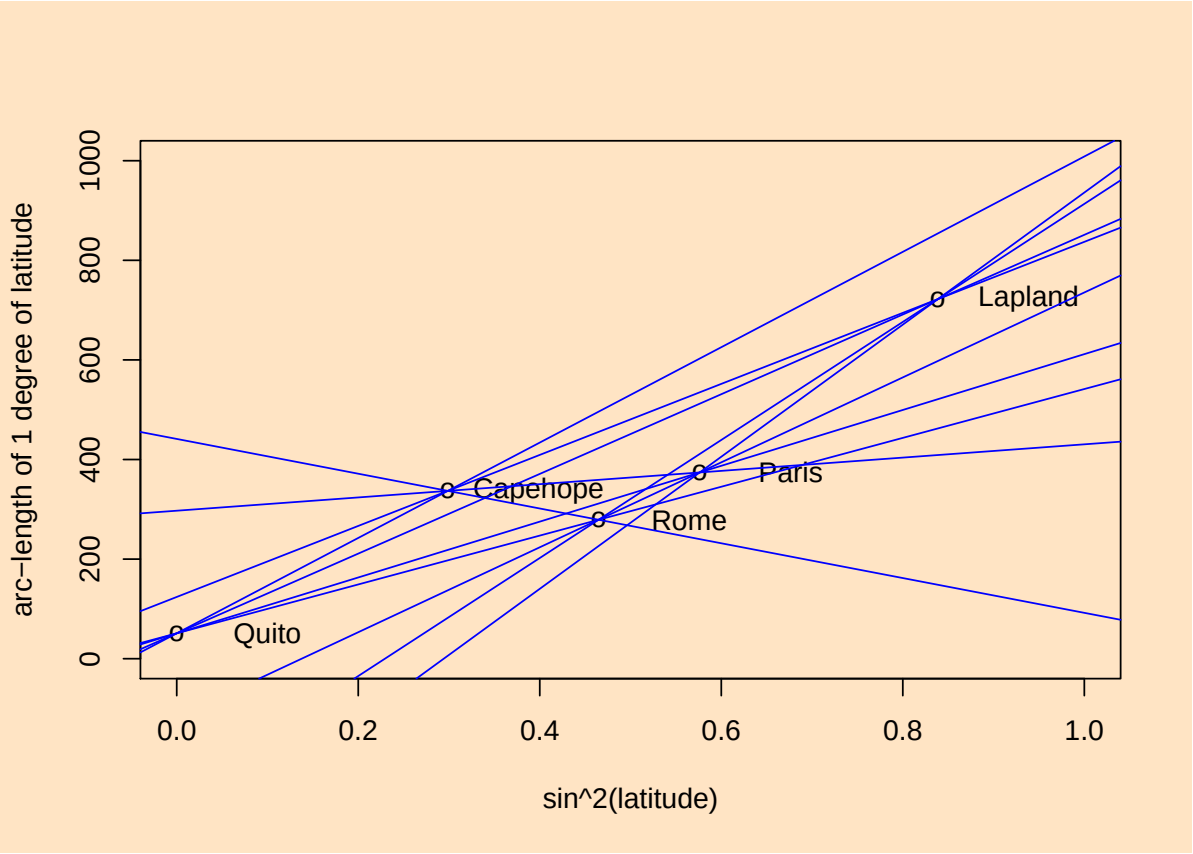
$$\sum_{i=1}^{j} w_{(i)} > \frac{1}{2} \sum_{i=1}^{n} w_{(i)}.$$

An Historical Prelude: The Ellipticity of the Earth



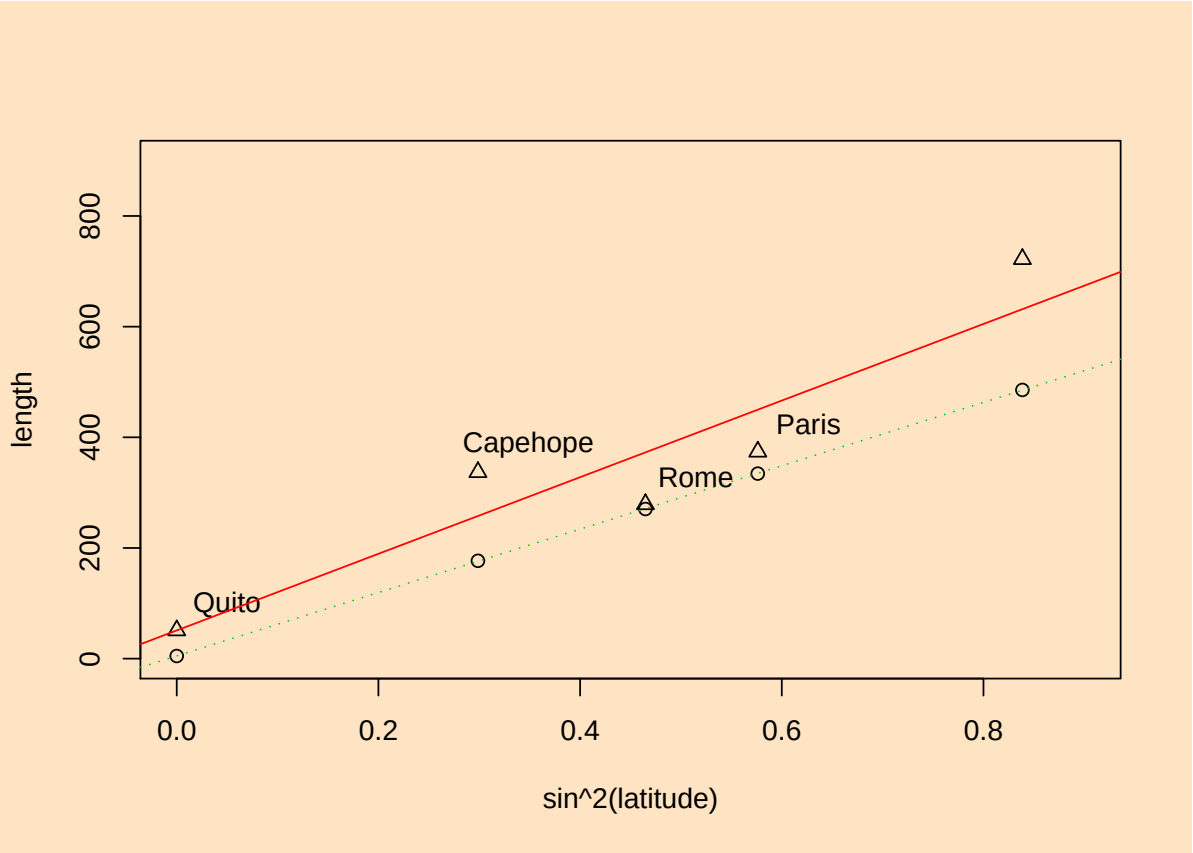
Grapefruit or Lemon? Boscovich (1755) considered the five measurements depicted above. Arc length is measured in toise per degree of latitude minus 56,700. An upward slope in this figure indicates that the earth is oblate (like a grapefruit) rather than prolate (like a lemon).

The First Regression



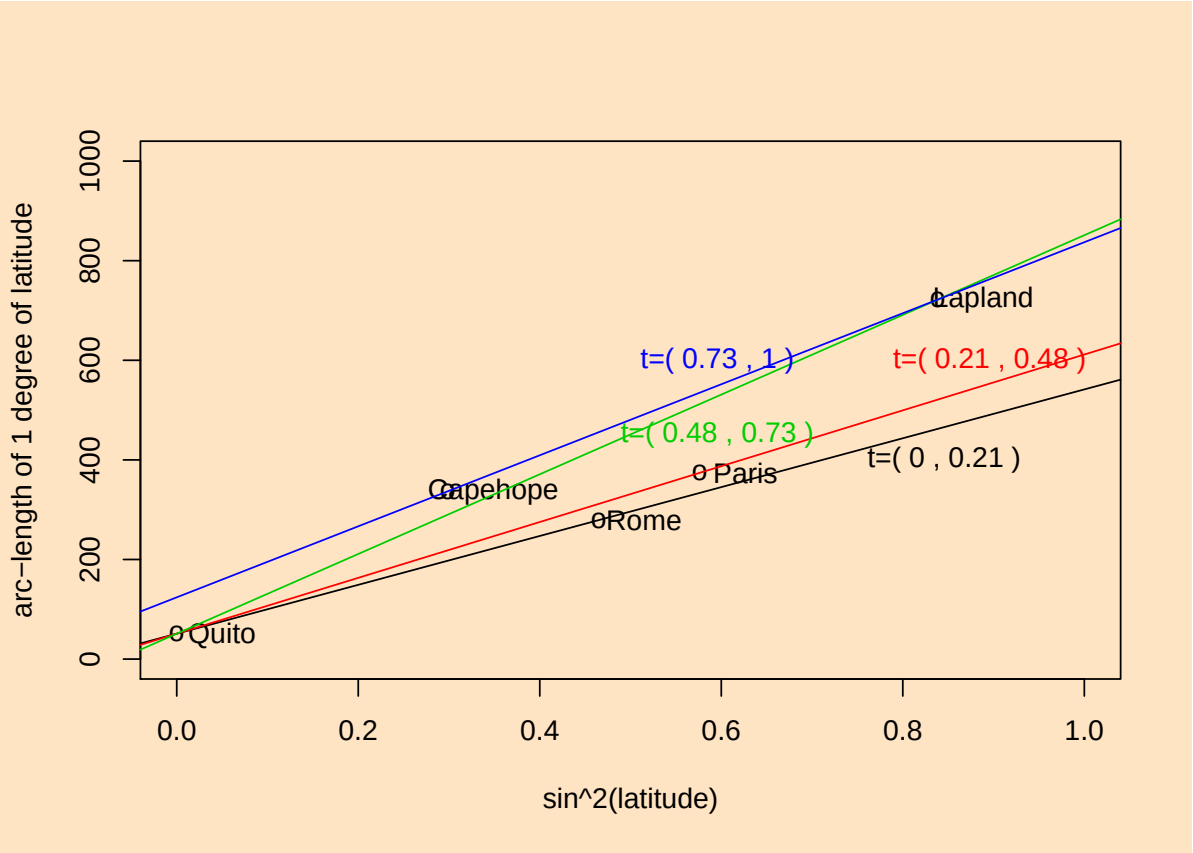
Boscovich (1755) computed all the pairwise slopes and initially reported a trimmed mean of the pairwise slopes as a point estimate of the earth's ellipticity.

The Second Regression



In 1757 Boscovich reestimated the model minimizing the sum of absolute residuals subject to the constraint that the residuals summed to zero to obtain the solid line fitting the triangular points. Modern measurements of the arc lengths are represented by the circles.

Earth's Ellipticity: A Quantile Regression View



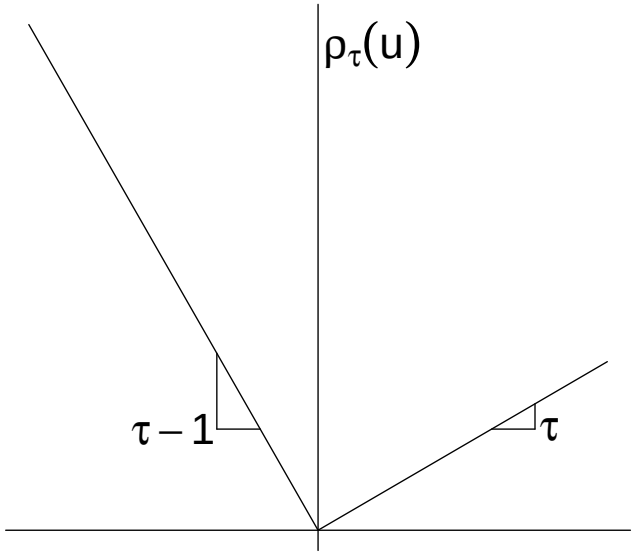
The quantile regression analysis of the Boscovich data identifies four distinct pairs of points that solve the weighted ℓ_1 problem for various intervals of the parameter τ.

Sample Quantiles via Optimization

The τth sample quantile can be defined as any solution to:

$$\hat{\alpha}(\tau) = \text{argmin}_{a \in \Re} \sum_{i=1}^{n} \rho_\tau(y_i - a)$$

where $\rho_\tau(u) = (\tau - I(u < 0))u$ as illustrated below.



The Least Squares Meta-Model

The unconditional mean solves

$$\mu = \min_{m} E(Y - m)^2$$

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Similarly, the unconditional τth quantile solves

$$\alpha_\tau = \min_a E\rho_\tau(Y - a)$$

and the conditional τth quantile solves

$$\alpha_\tau(x) = \min_a E_{Y|X=x}\rho_\tau(Y - a(X))$$

Regression Quantiles via Optimization

The sample analogue of the foregoing population concepts yields, the nonparametric quantile regression estimator

$$\hat{\alpha}_\tau(x) = \text{argmin}_{a \in \mathcal{A}} \sum_{i=1}^{n} \rho_\tau(y_i - a(x_i))$$

If we take $\mathcal{A} = \{a : \mathbb{R}^p \to \mathbb{R} | a(x) = x^\top \beta,\ \beta \in \mathbb{R}^p\}$, then we have the linear (in parameters) quantile regression problem:

$$\hat{\beta}(\tau) = \text{argmin}_{b \in \mathbb{R}} \sum_{i=1}^{n} \rho_\tau(y_i - x_i^\top b)$$

Beyond Average Treatment Effects

Lehmann (1974) proposed the following general model of treatment response:

> *"Suppose the treatment adds the amount $\Delta(x)$ when the response of the untreated subject would be x. Then the distribution G of the treatment responses is that of the random variable $X + \Delta(X)$ where X is distributed according to F."*

Lehmann QTE as a QQ-Plot

Doksum (1974) defines $\Delta(x)$ as the "horizontal distance" between F and G at x, *i.e.*

$$F(x) = G(x + \Delta(x)).$$

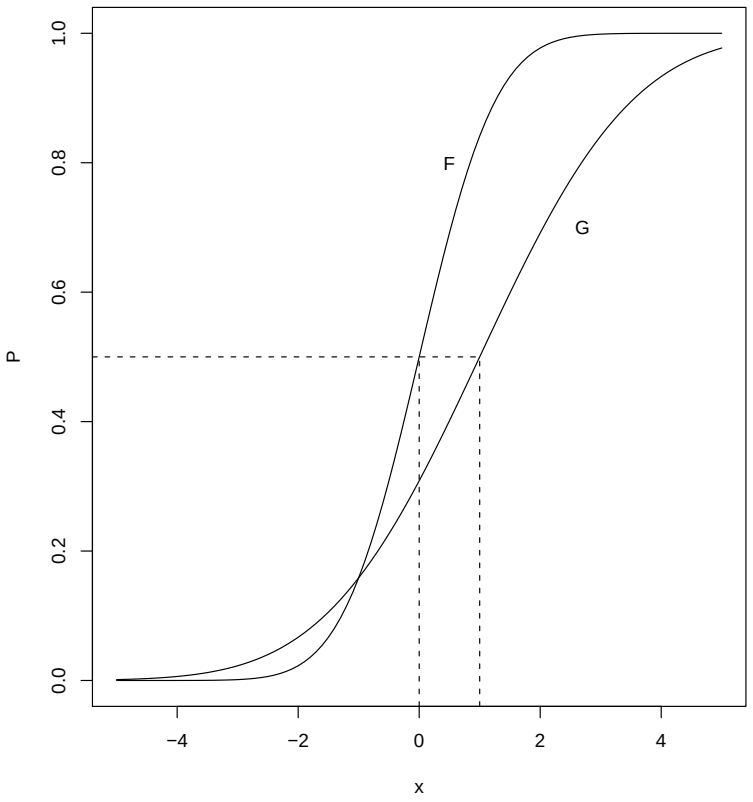
Then $\Delta(x)$ is uniquely defined as

$$\Delta(x) = G^{-1}(F(x)) - x.$$

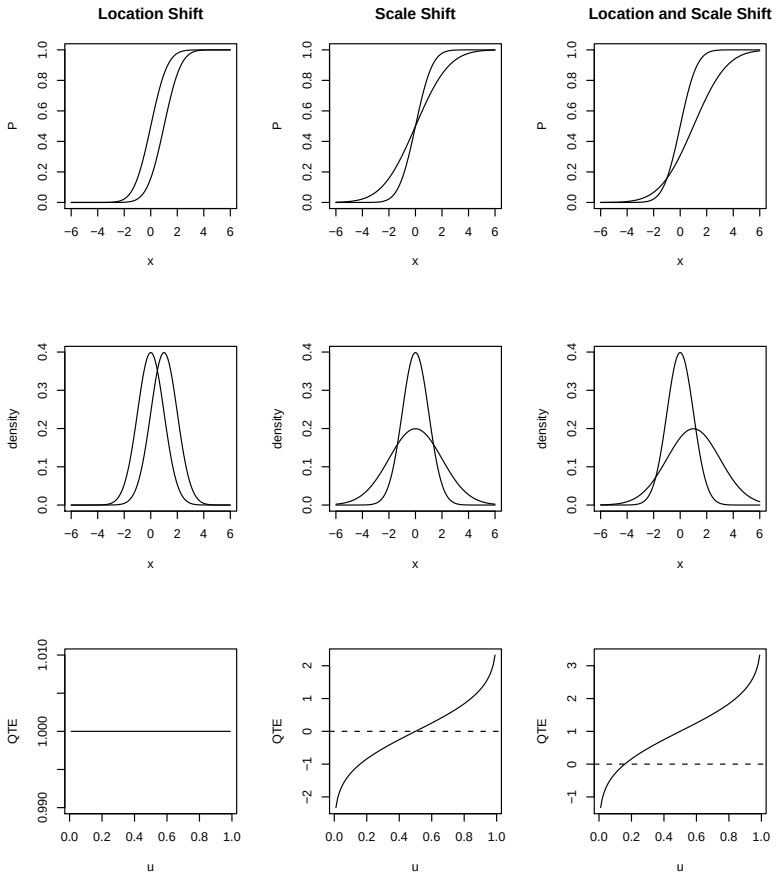
This is the essence of the conventional QQ-plot. Changing variables so $\tau = F(x)$ we have the quantile treatment effect (QTE):

$$\delta(\tau) = \Delta(F^{-1}(\tau)) = G^{-1}(\tau) - F^{-1}(\tau).$$

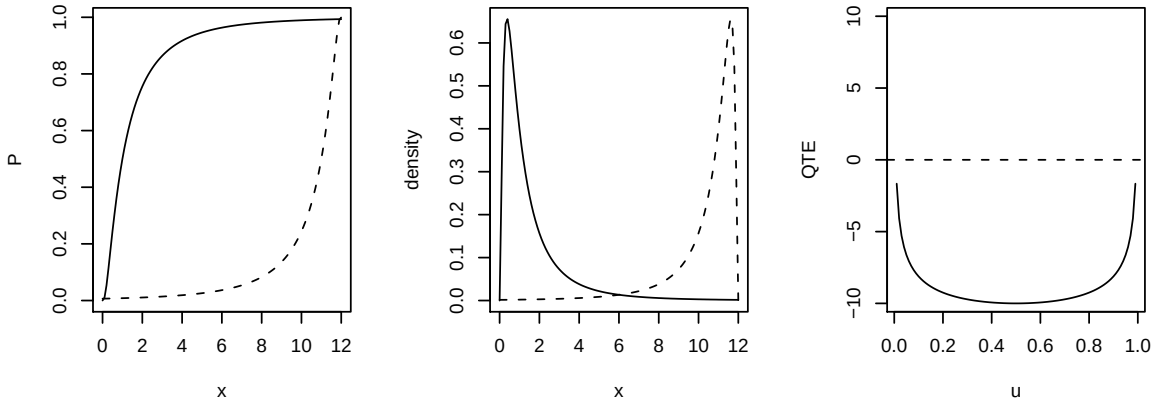
Lehmann-Doksum QTE



Lehmann-Doksum QTE



An Asymmetric Example



Treatment shifts the distribution from right skewed to left skewed making the QTE U-shaped.

The Erotic is Unidentified

The Lehmann QTE characterizes the difference in the marginal distributions, F and G, but it cannot reveal anything about the joint distribution, H. The copula function, Schweizer and Wolf (1981), Genest and McKay, (1986),

$$\varphi(u, v) = H(F^{-1}(u), G^{-1}(v)),$$

is *not* identified.

QTE via Quantile Regression

The Lehmann QTE is naturally estimable by

$$\hat{\delta}(\tau) = \hat{G}_n^{-1}(\tau) - \hat{F}_m^{-1}(\tau)$$

where $\hat{G}_n$ and $\hat{F}_m$ denote the empirical distribution functions of the treatment and control observations, Consider the quantile regression model

$$Q_{Y_i}(\tau|D_i) = \alpha(\tau) + \delta(\tau)D_i$$

where D_i denotes the treatment indicator, and $Y_i = h(T_i)$, *e.g.* $Y_i = \log T_i$, which can be estimated by solving,

$$\min \sum_{i=1}^{n} \rho_\tau(y_i - \alpha - \delta D_i)$$

Computation of Quantile Regression

Primal Formulation as a Linear Program

$$\min\{\tau 1^\top u + (1-\tau)1^\top v | y = Xb + u - v, (b, u, v) \in \mathbb{R}^p \times \mathbb{R}^{2n}_+\}$$

Dual Formulation as a Linear Program

$$\max\{y'd | X^\top d = (1-\tau)X^\top 1, d \in [0,1]^n\}$$

Solutions are characterized by an exact fit to p observations.

Equivariance of Regression Quantiles

- Scale Equivariance: For any $a > 0$, $\hat{\beta}(\tau; ay, X) = a\hat{\beta}(\tau; y, X)$ and $\hat{\beta}(\tau; -ay, X) = a\hat{\beta}(1 - \tau; y, X)$

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- Regression Shift: For any $\gamma \in \mathbb{R}^p$ $\hat{\beta}(\tau; y + X\gamma, X) = \hat{\beta}(\tau; y, X) + \gamma$
- Reparameterization of Design: For any $|A| \neq 0$, $\hat{\beta}(\tau; y, AX) = A^{-1}\hat{\beta}(\tau; yX)$

Equivariance to Monotone Transformations

For any monotone function h, conditional quantile functions $Q_Y(\tau|x)$ are equivariant in the sense that

$$Q_{h(Y)|X}(\tau|x) = h(Q_{Y|X}(\tau|x))$$

In contrast to conditional mean functions for which

$$E(h(Y)|X) \neq h(EY|X)$$

Examples:
$h(y) = \min\{0, y\}$, Powell(1985)
$h(y) = \text{sgn}\{y\}$ Rosenblatt(1957) Manski(1975)

Robustness

- Bounded Influence Function in y for fixed x_i, decent breakdown behavior for fixed design.
- Only the signs of the residuals $\hat{u} = y - X\hat{\beta}(\tau, y, X)$. matter

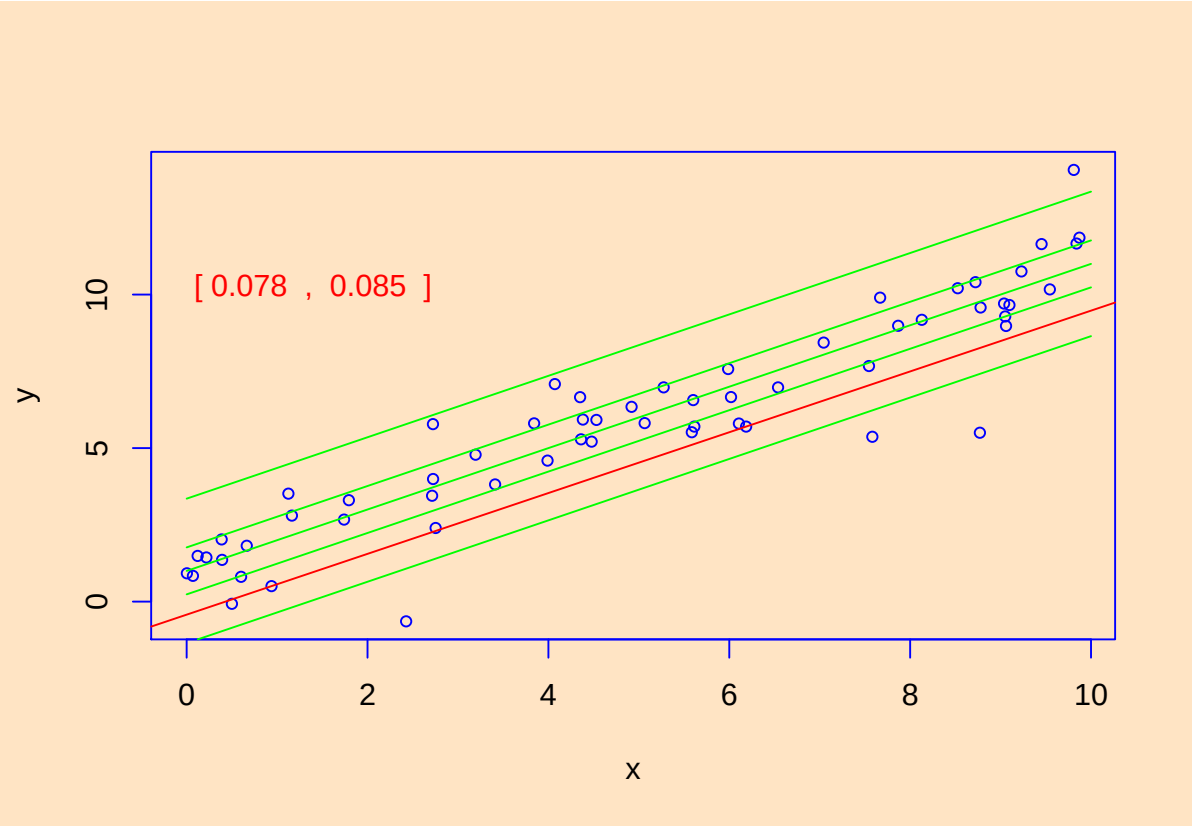
$$\hat{\beta}(\tau; y, X) = \hat{\beta}(\tau, y + D\hat{u}, X)$$

  for any diagonal matrix D with nonnegative elements.
- Robustness with respect to influential x observations is more challenging, but there are several very interesting proposals.

Quantile Regression: The Movie

- Bivariate linear model with iid Student t errors
- Conditional quantile functions are parallel in green
- 100 observations indicated in blue
- Fitted quantile regression lines in red

Quantile Regression in the iid Error Model



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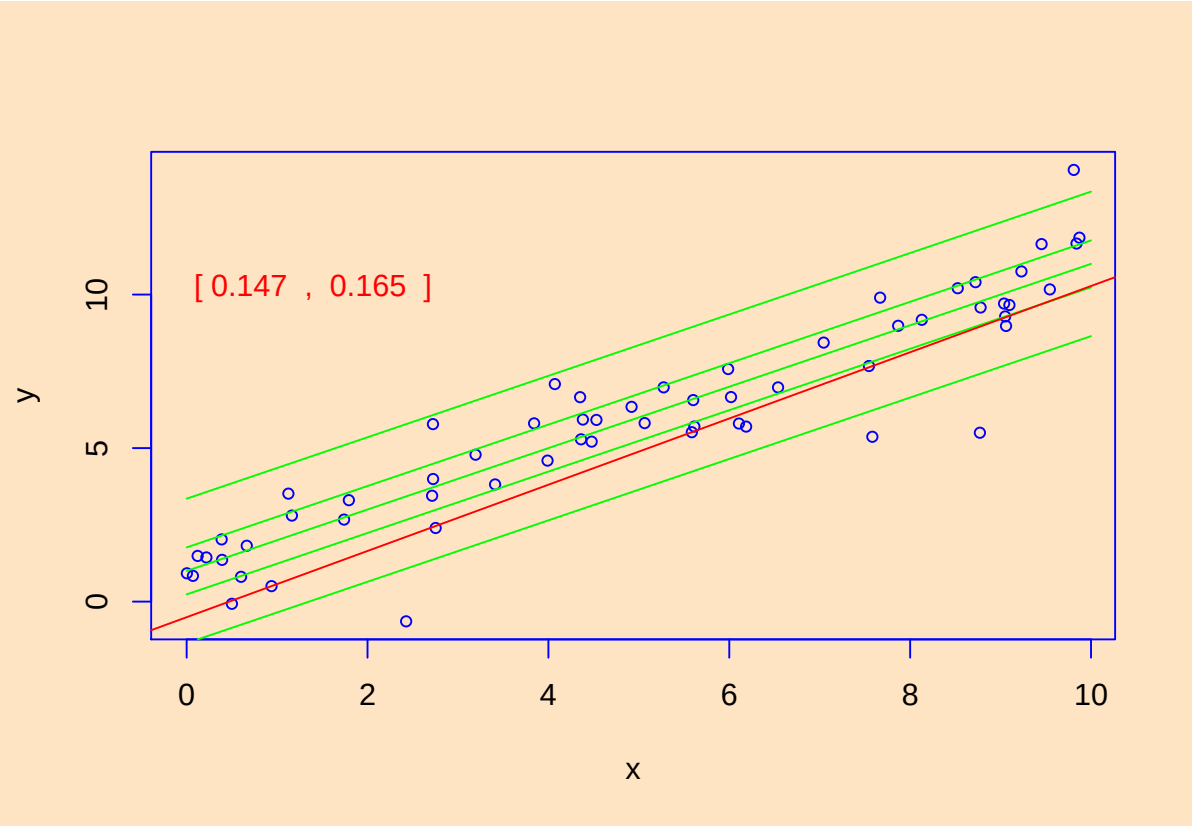

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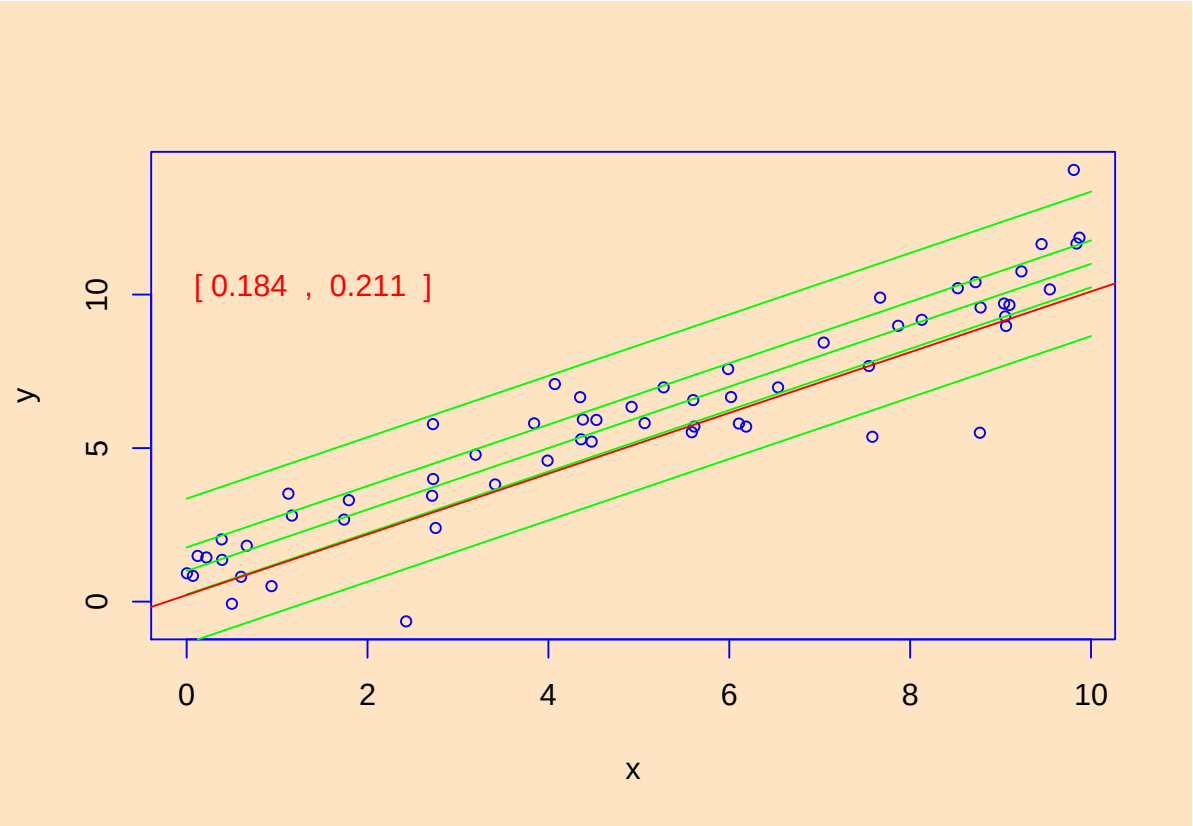

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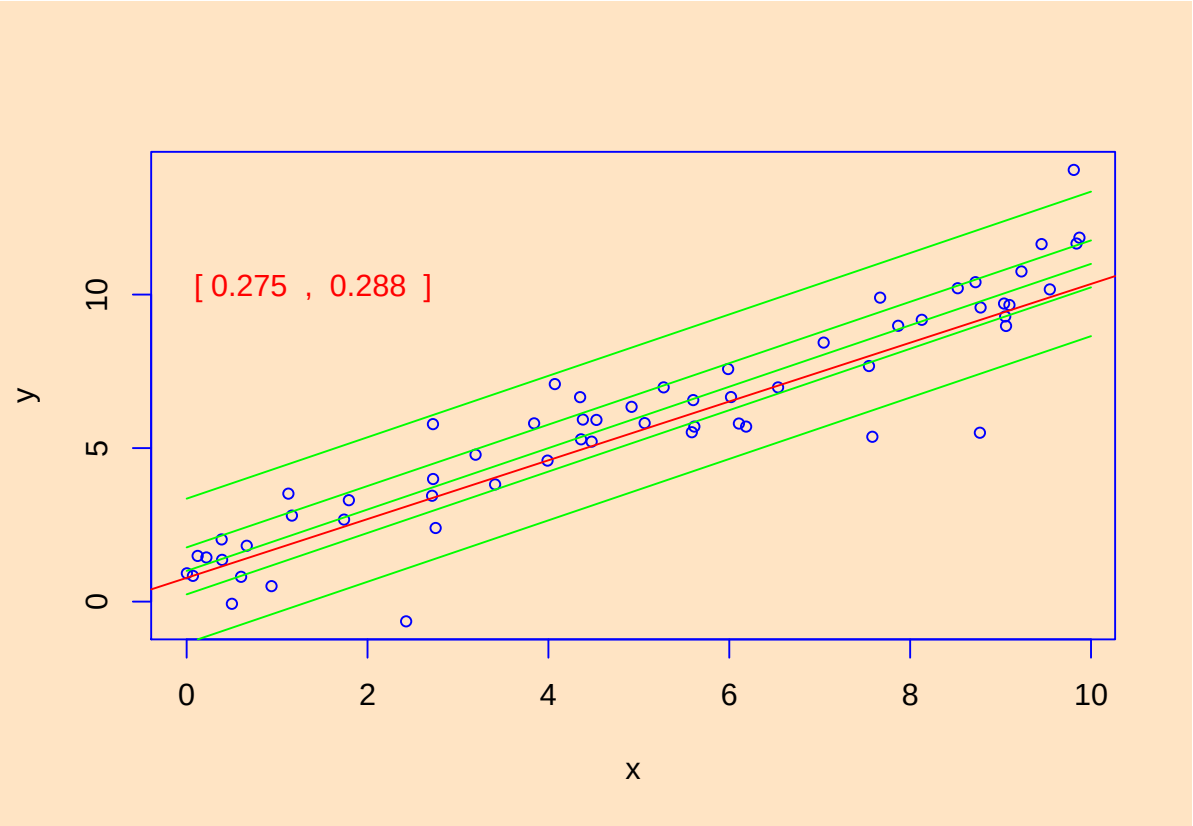

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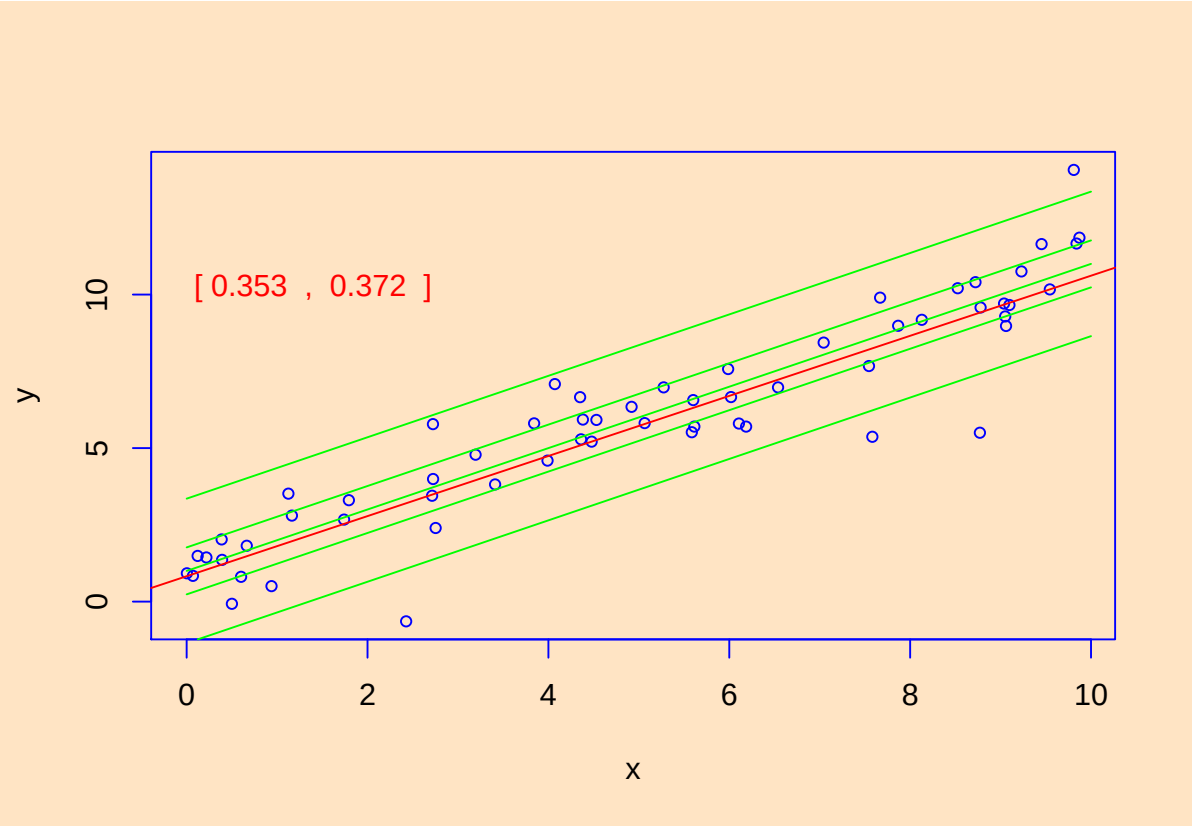

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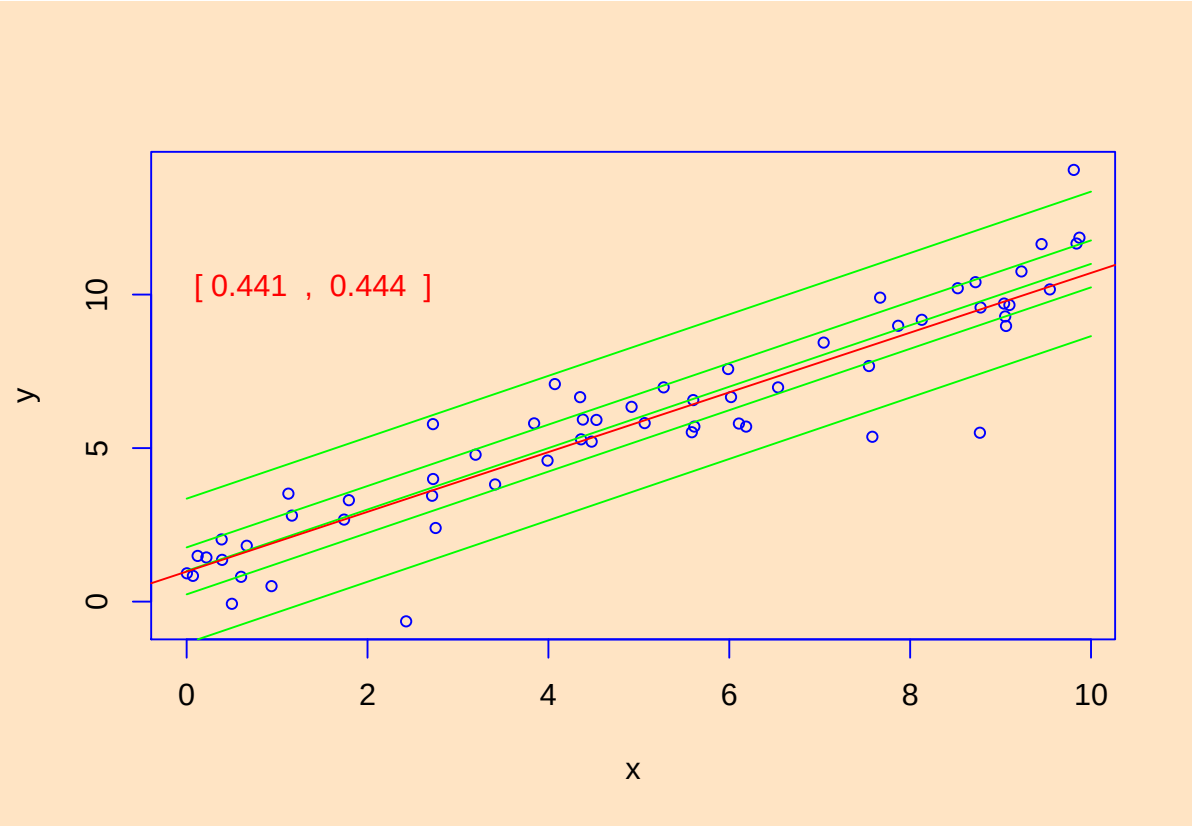

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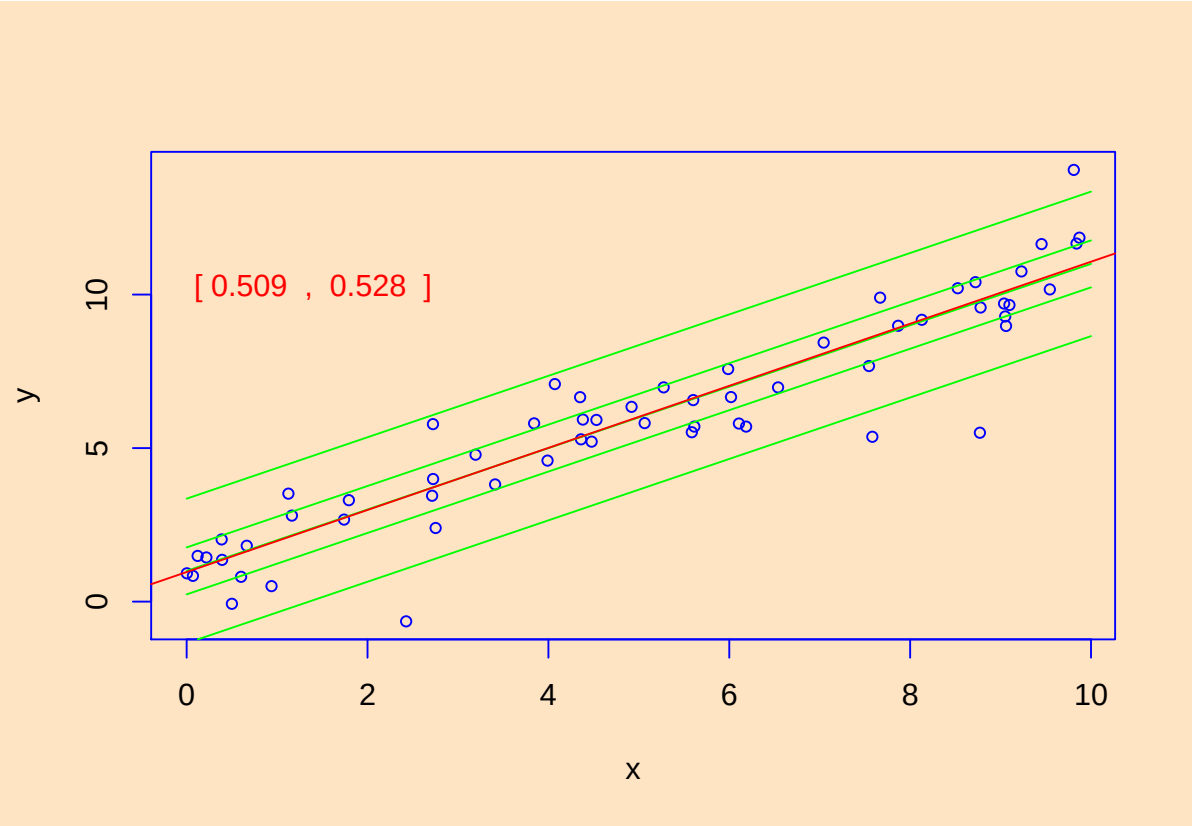

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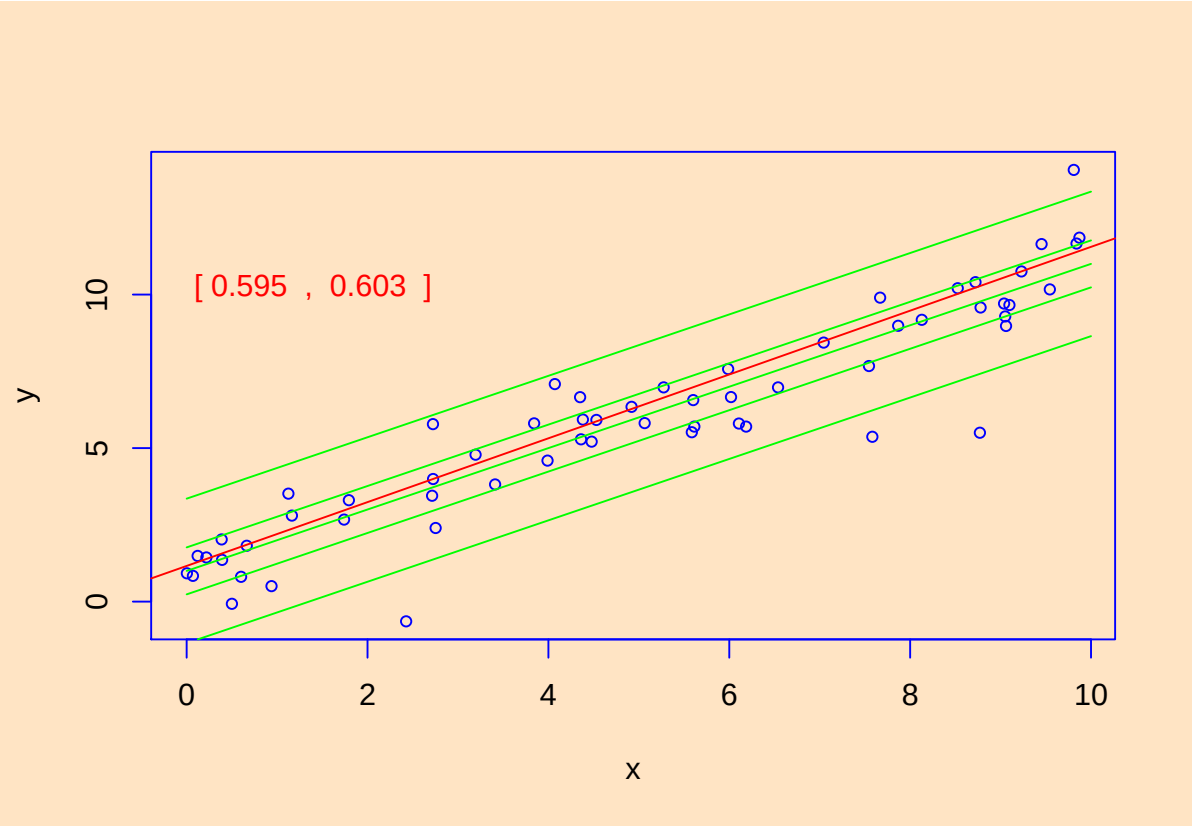

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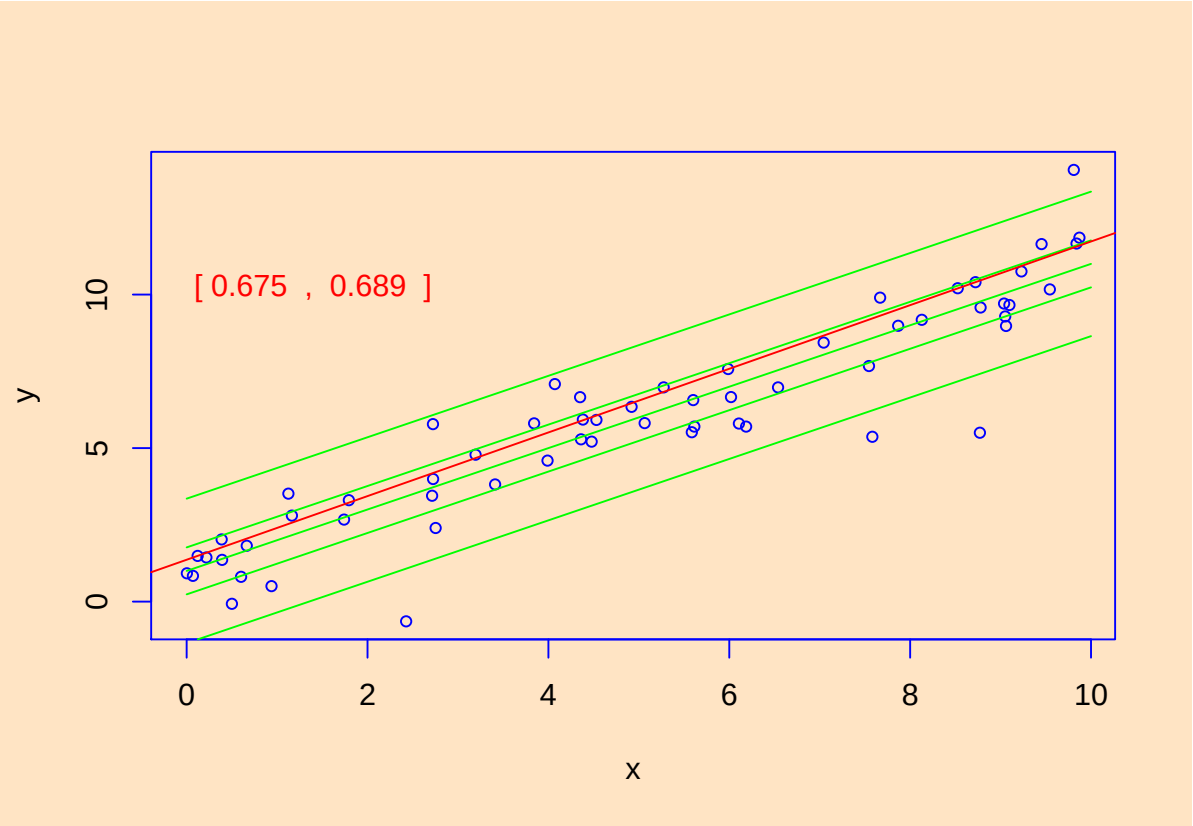

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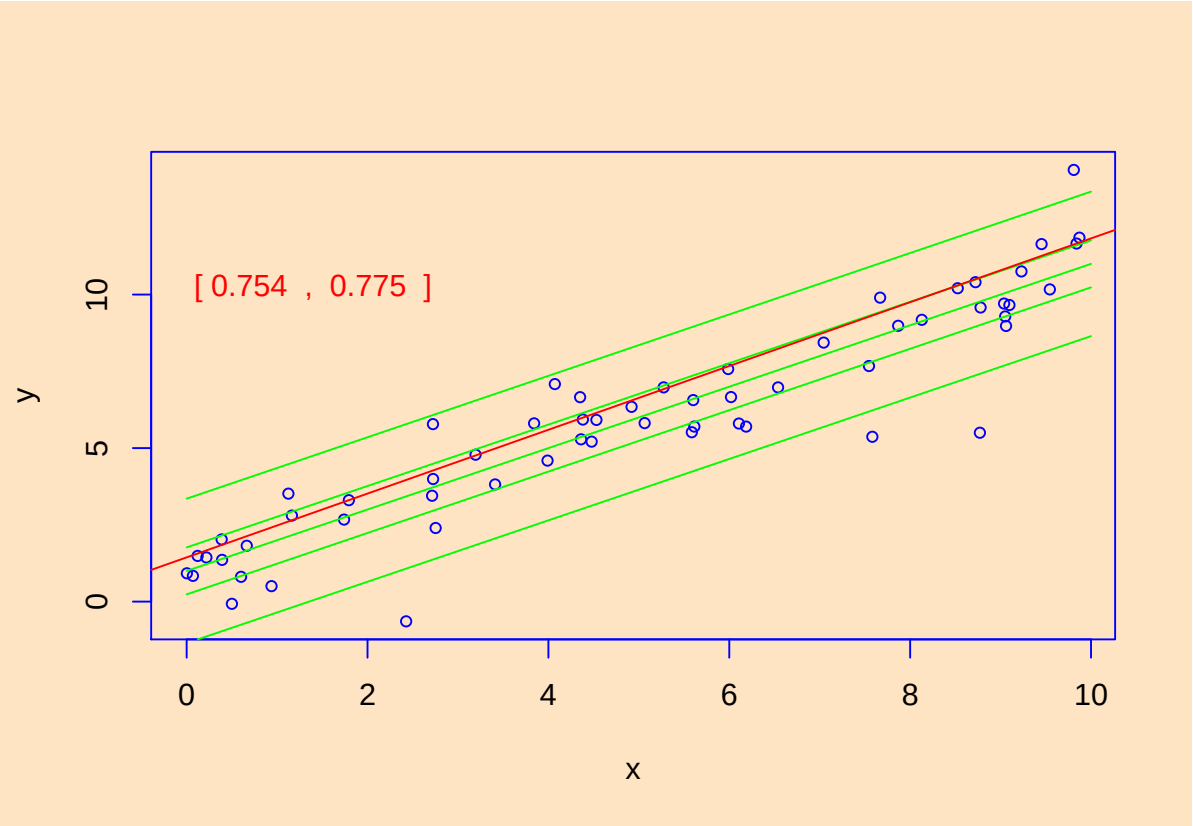

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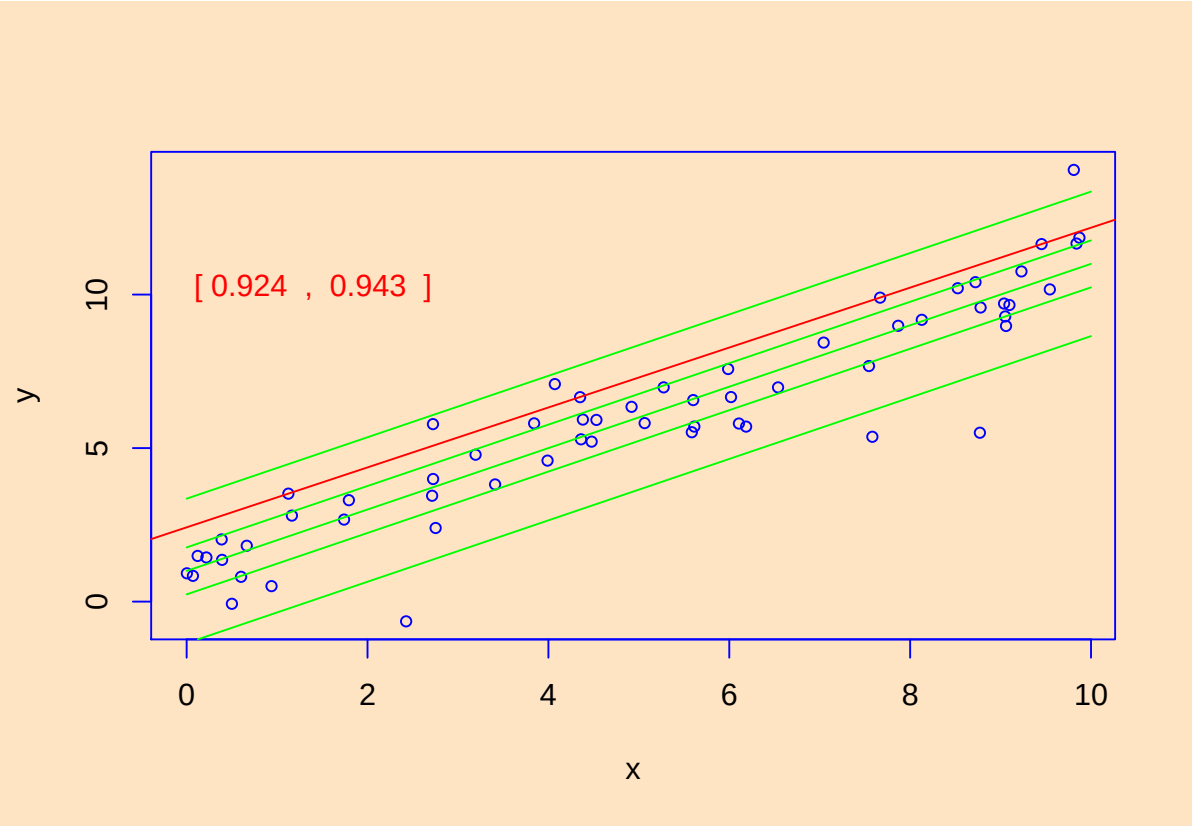
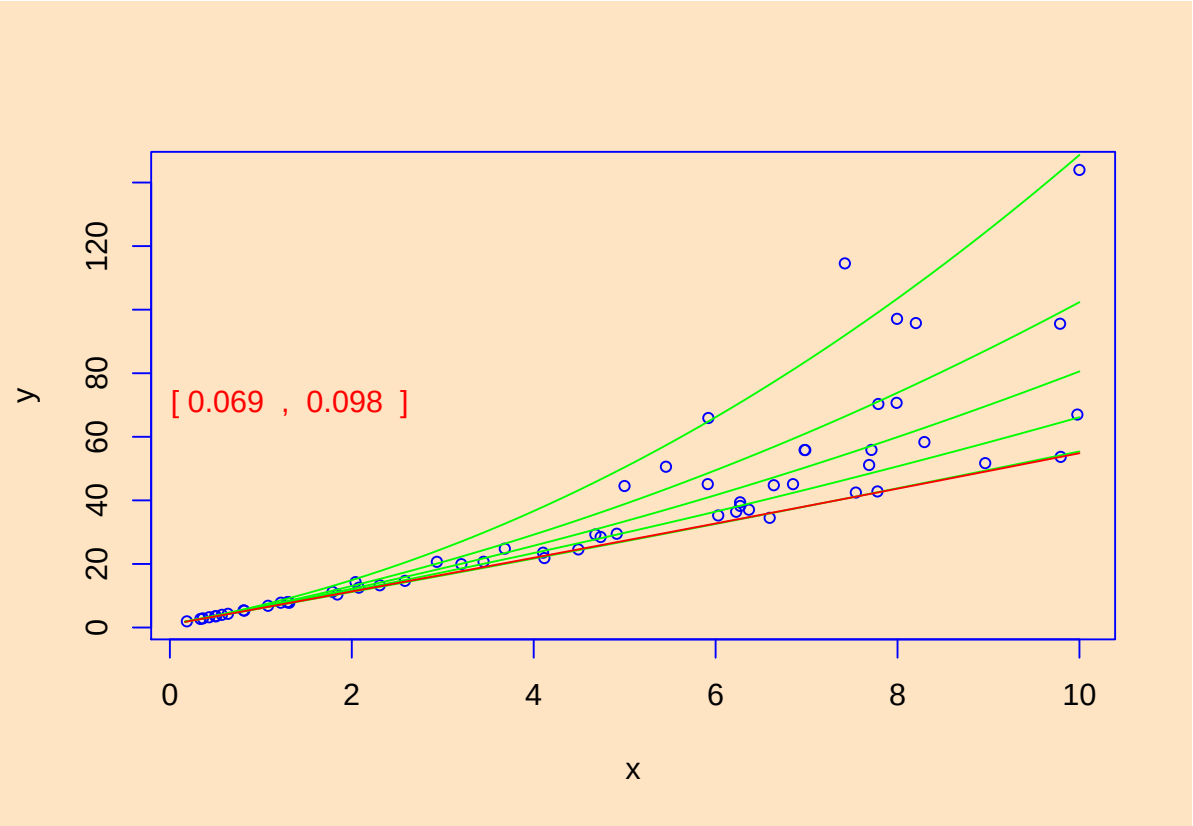


Virtual Quantile Regression II

- Bivariate quadratic model with Heteroscedastic χ^2 errors
- Conditional quantile functions drawn in green
- 100 observations indicated in blue
- Fitted quadratic quantile regression lines in red

Quantile Regression in the Heteroscedastic Error Model



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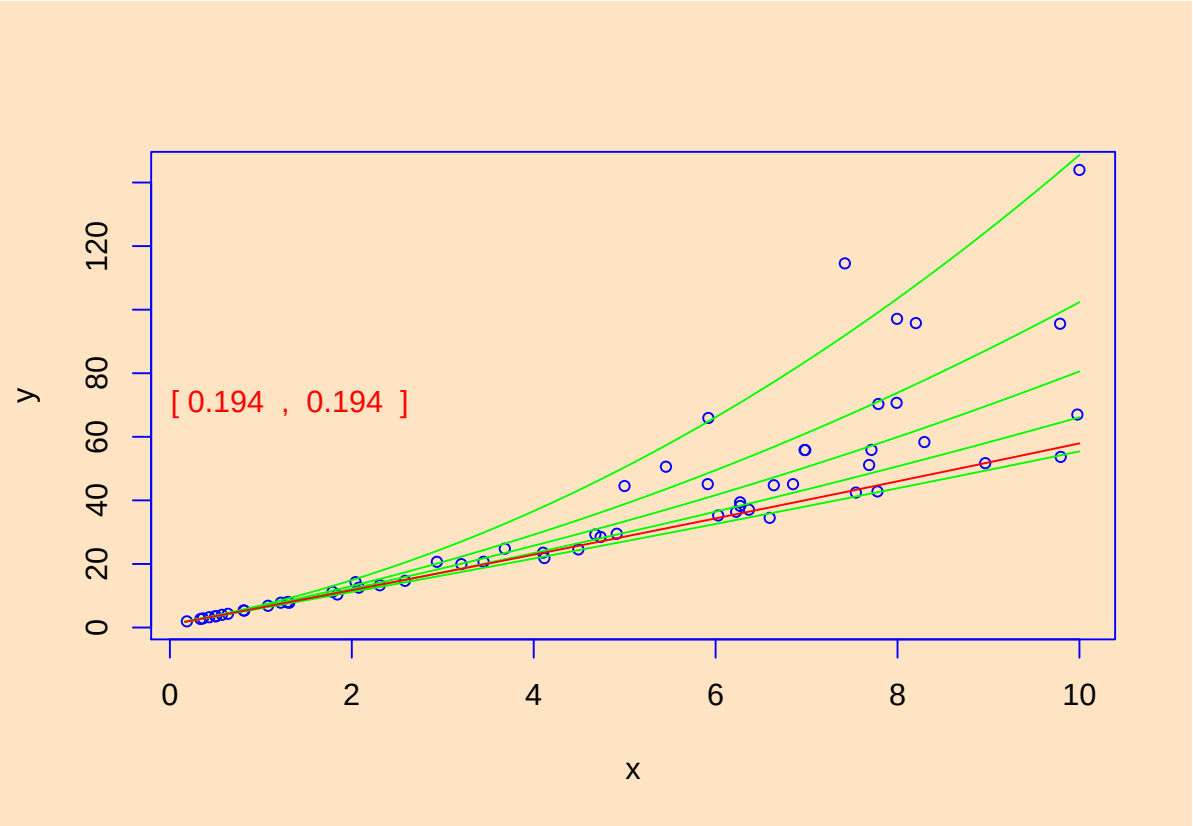

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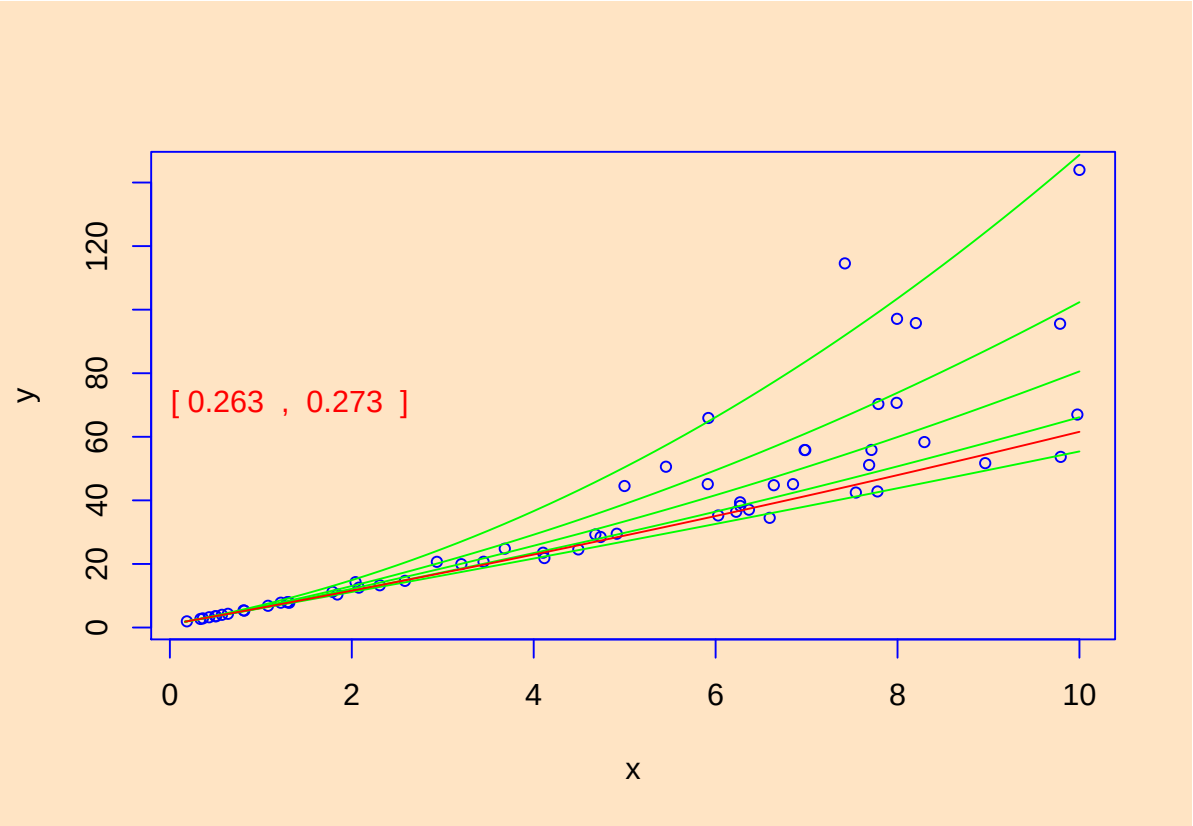

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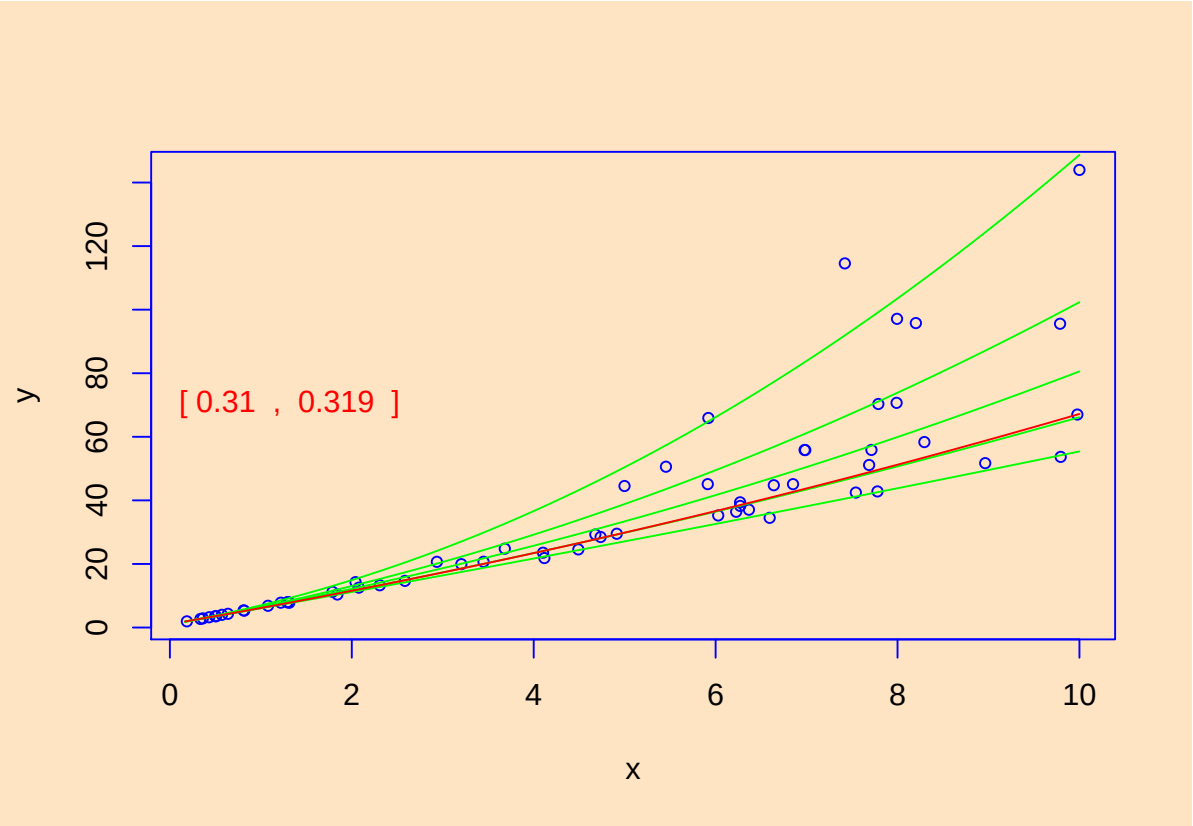

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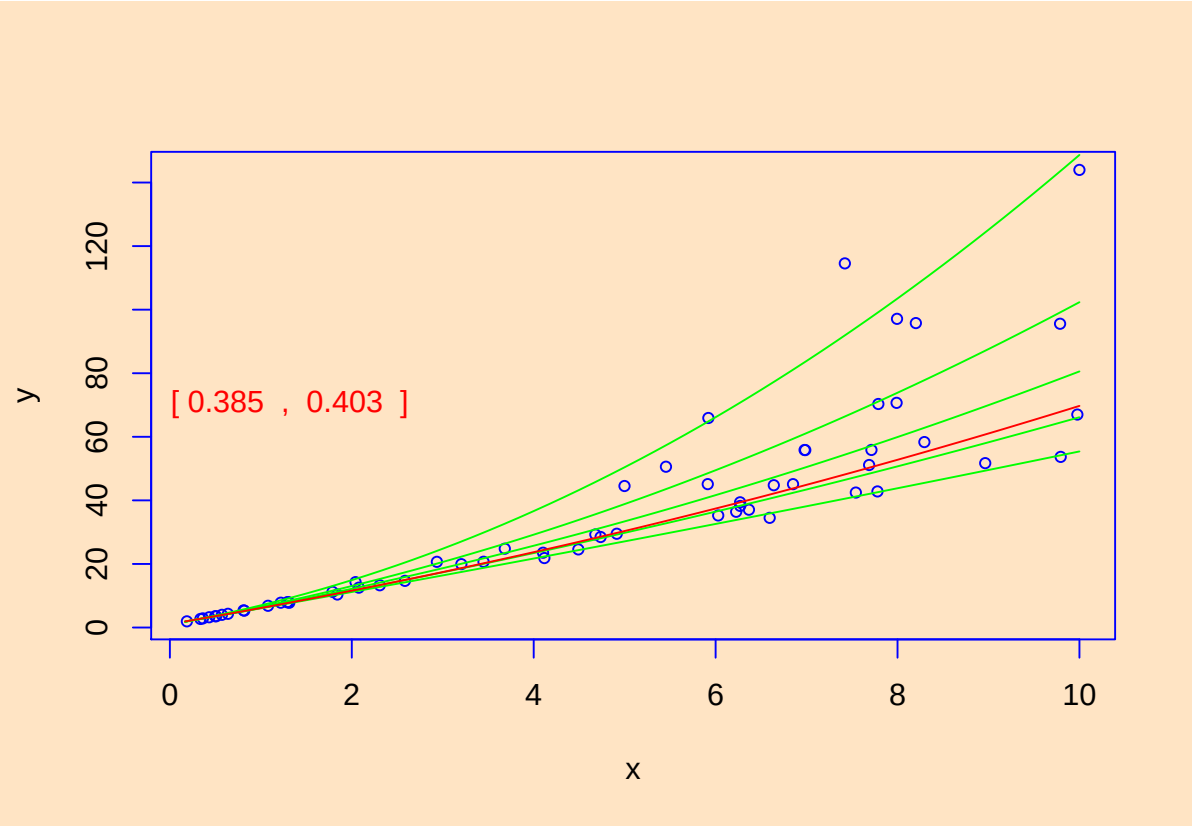

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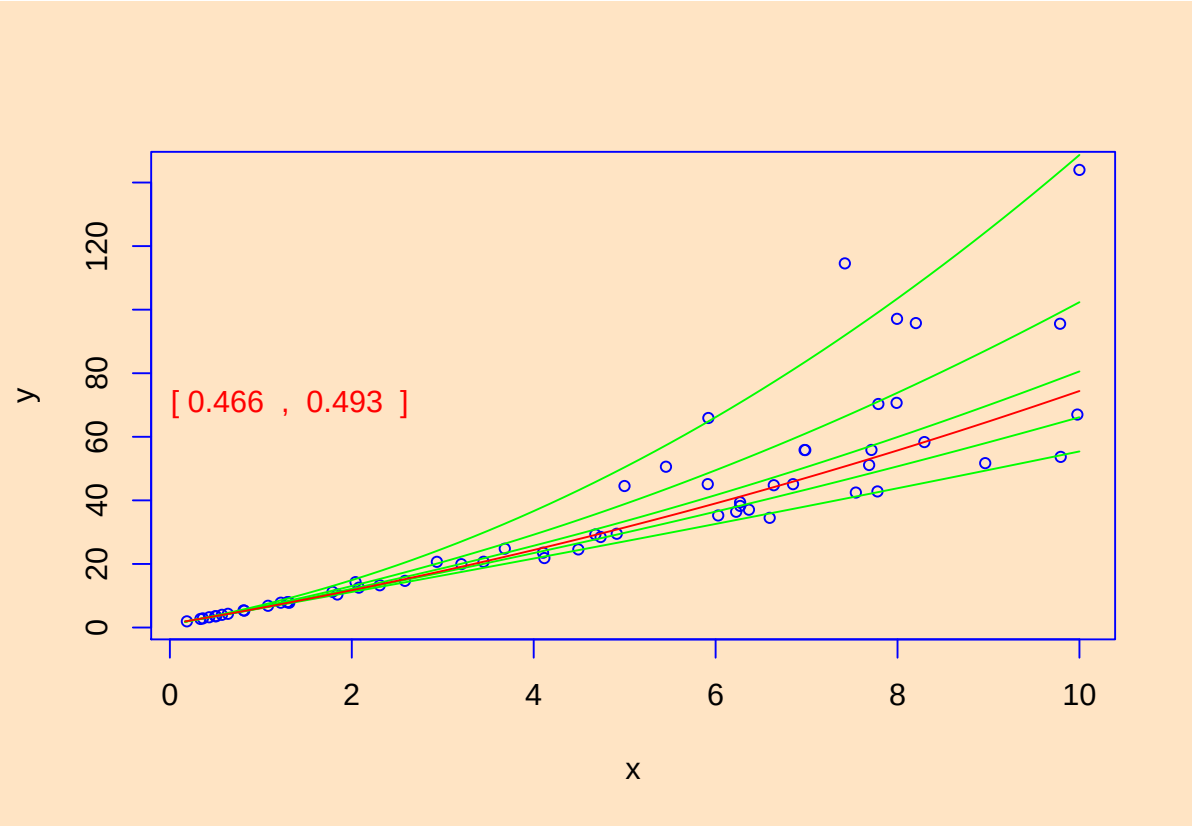

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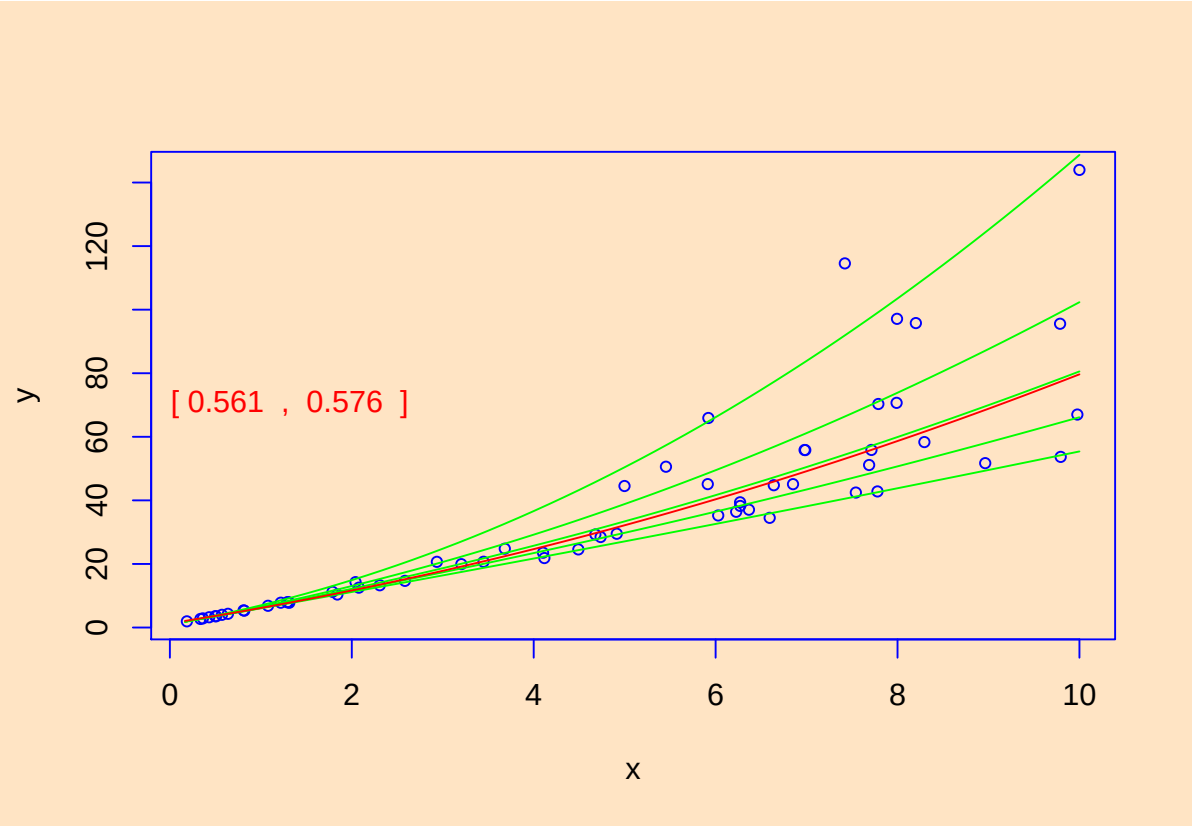

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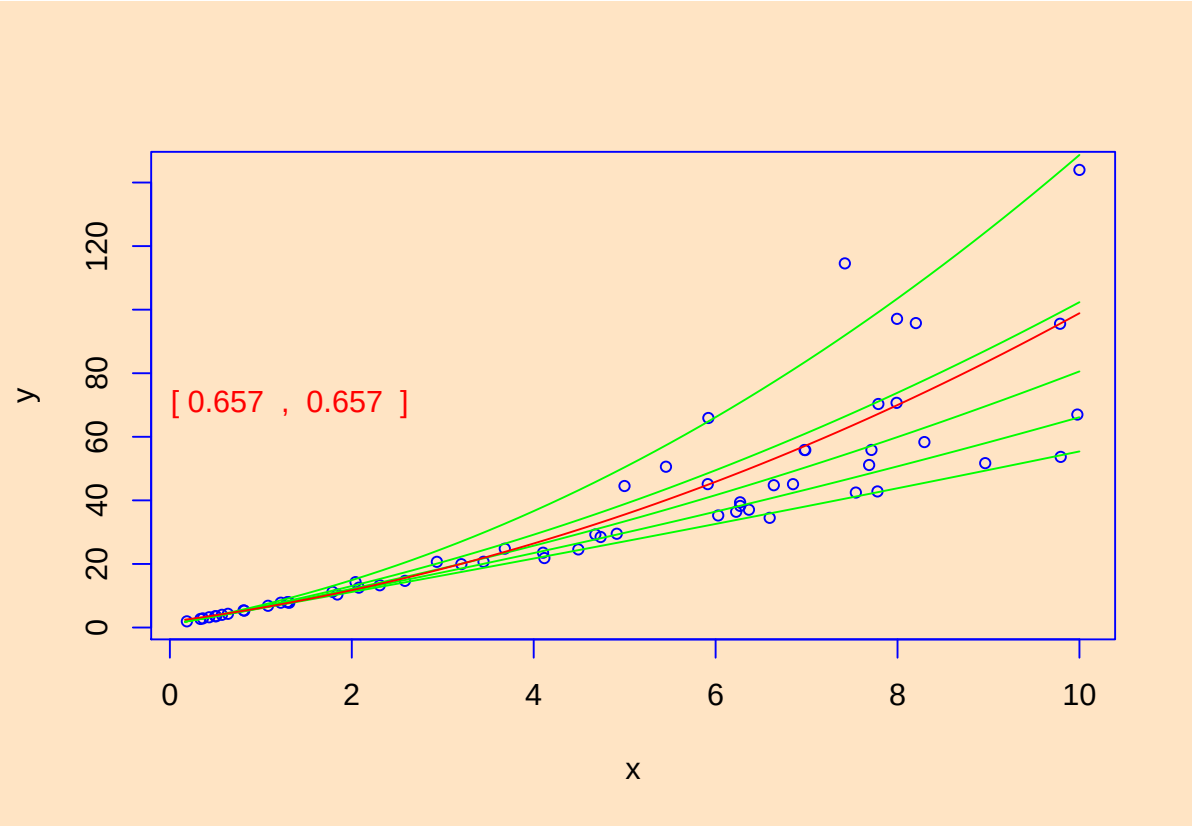

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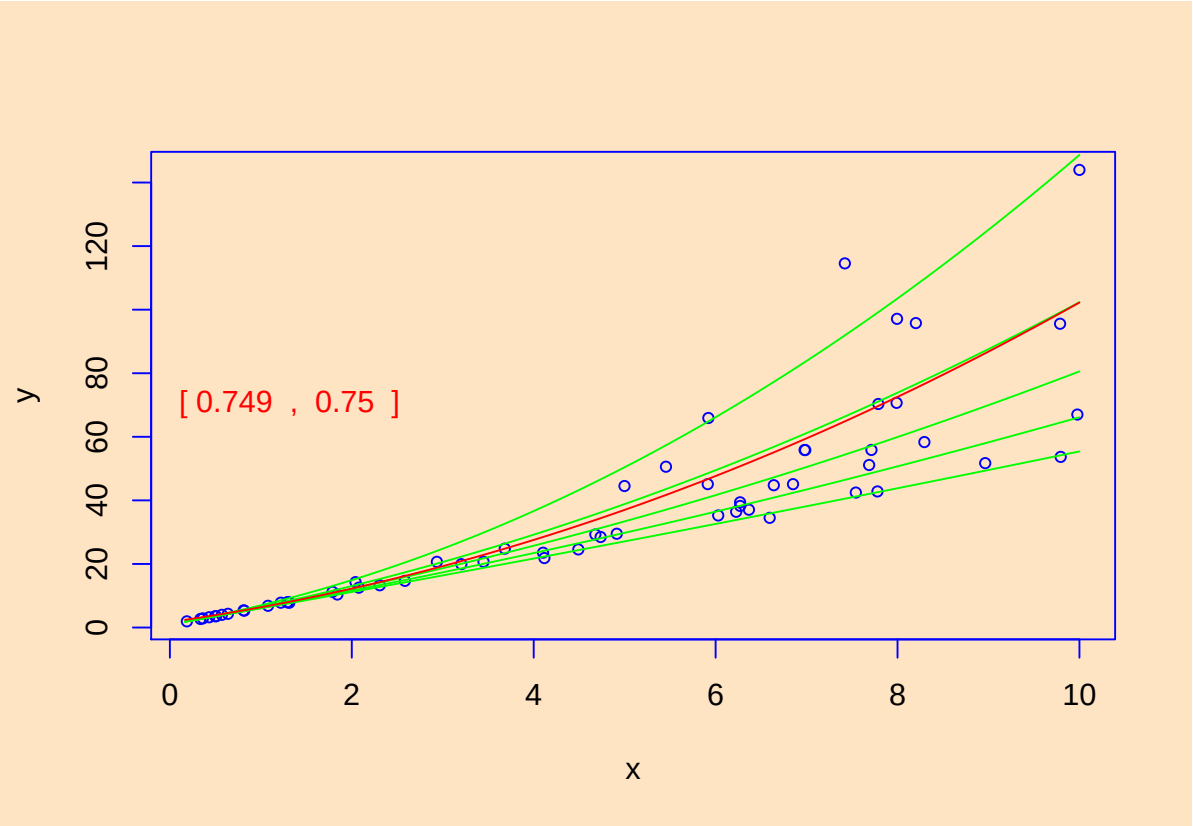

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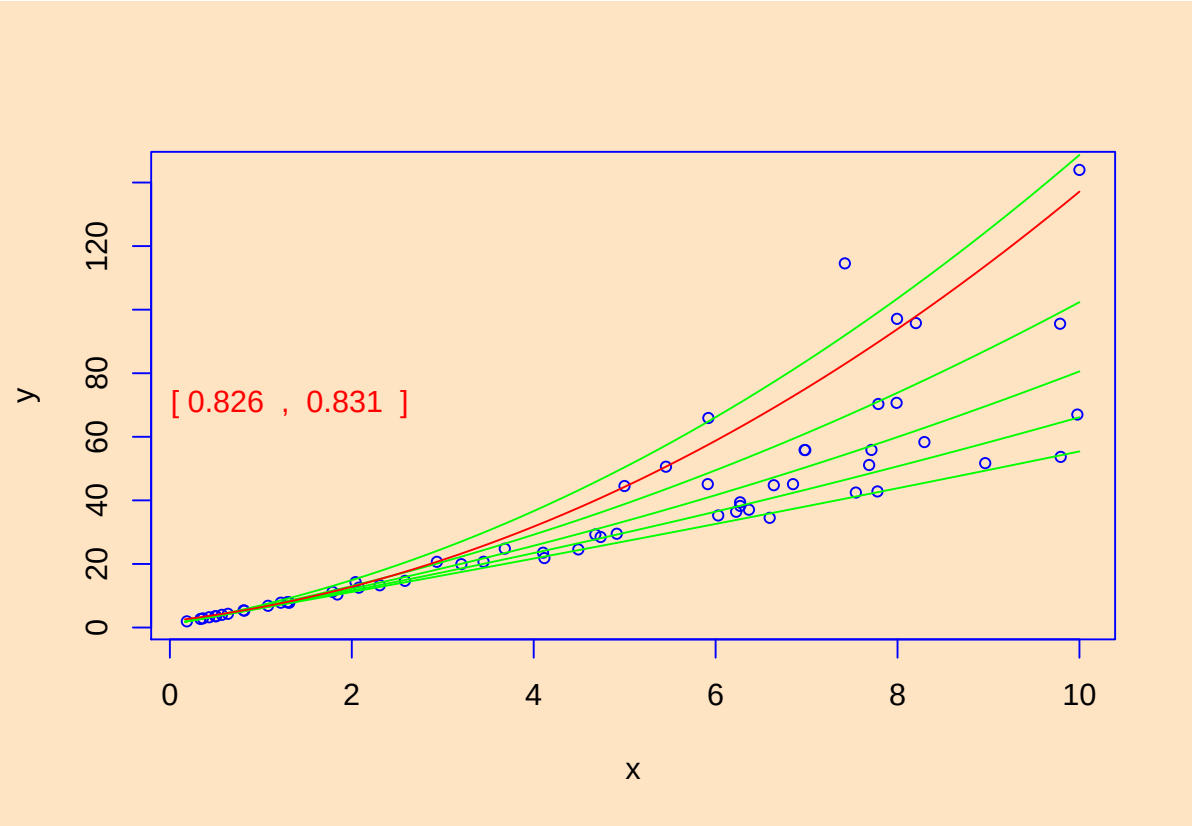

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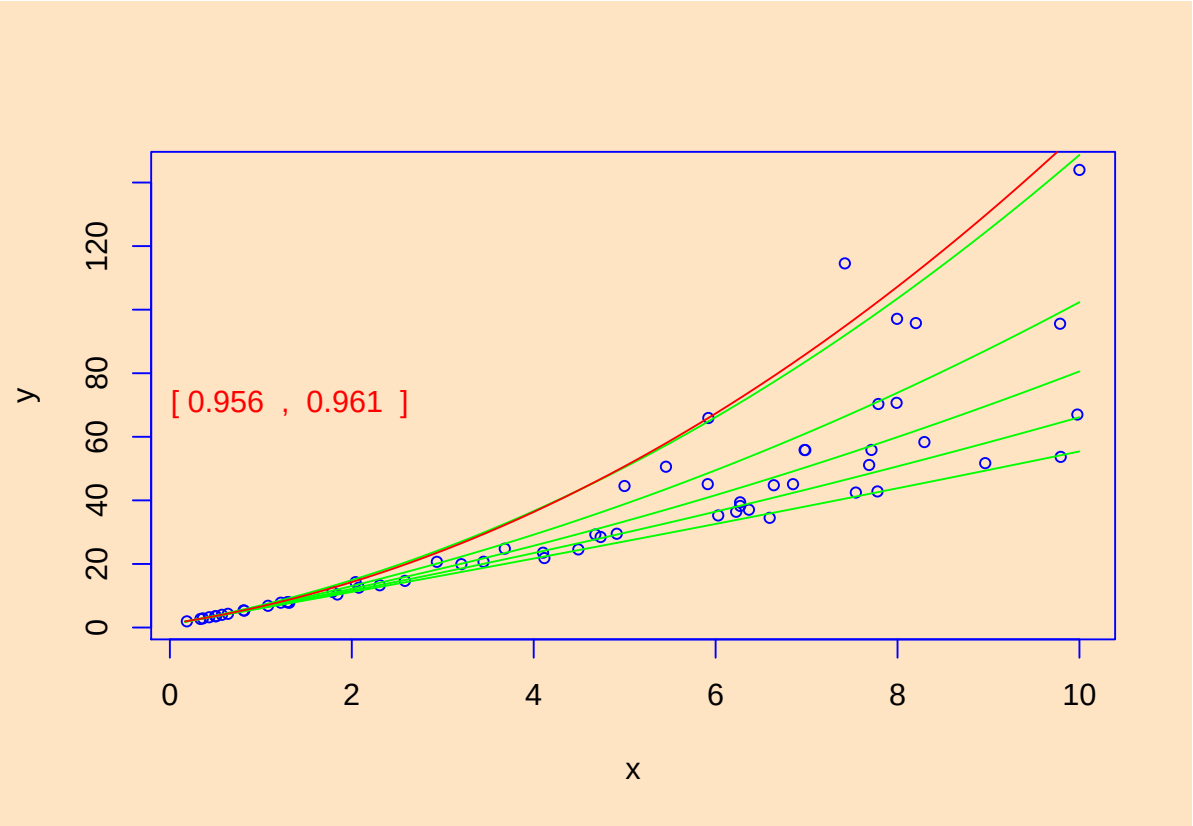
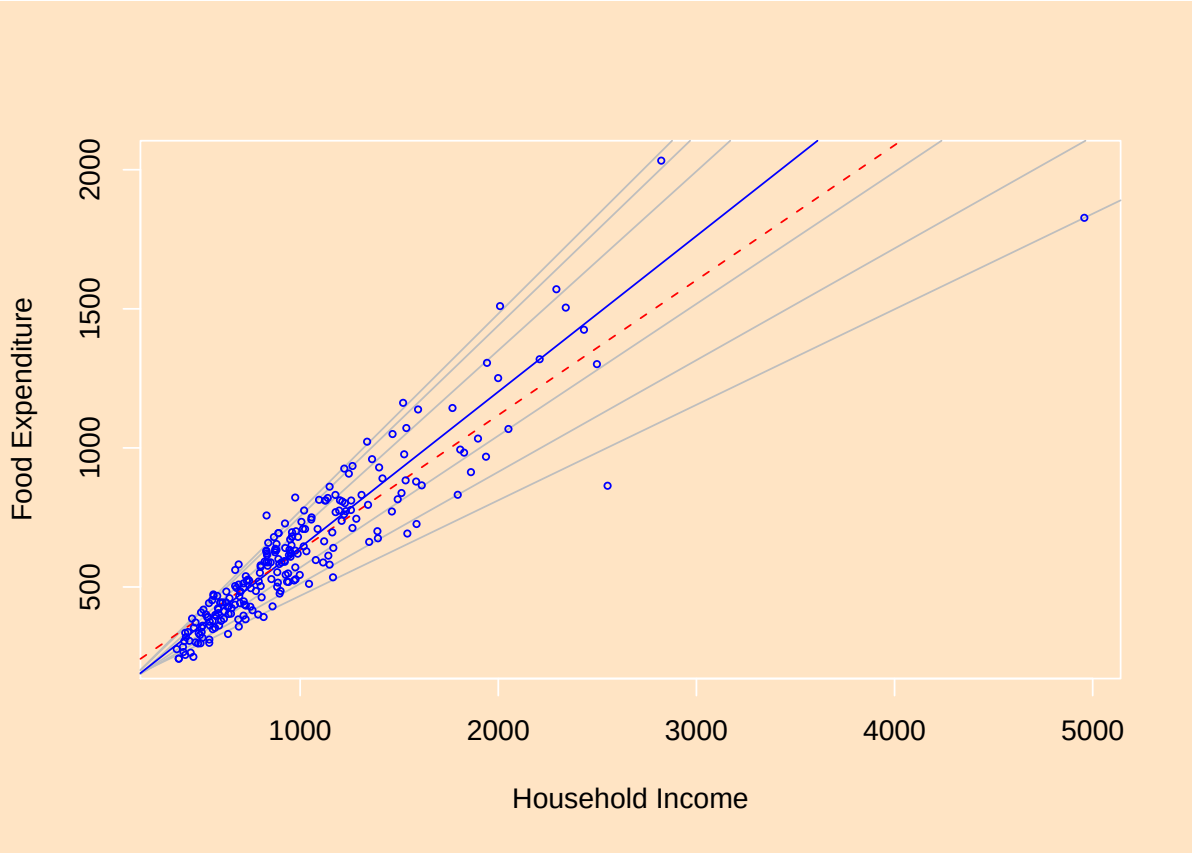


Three Applications

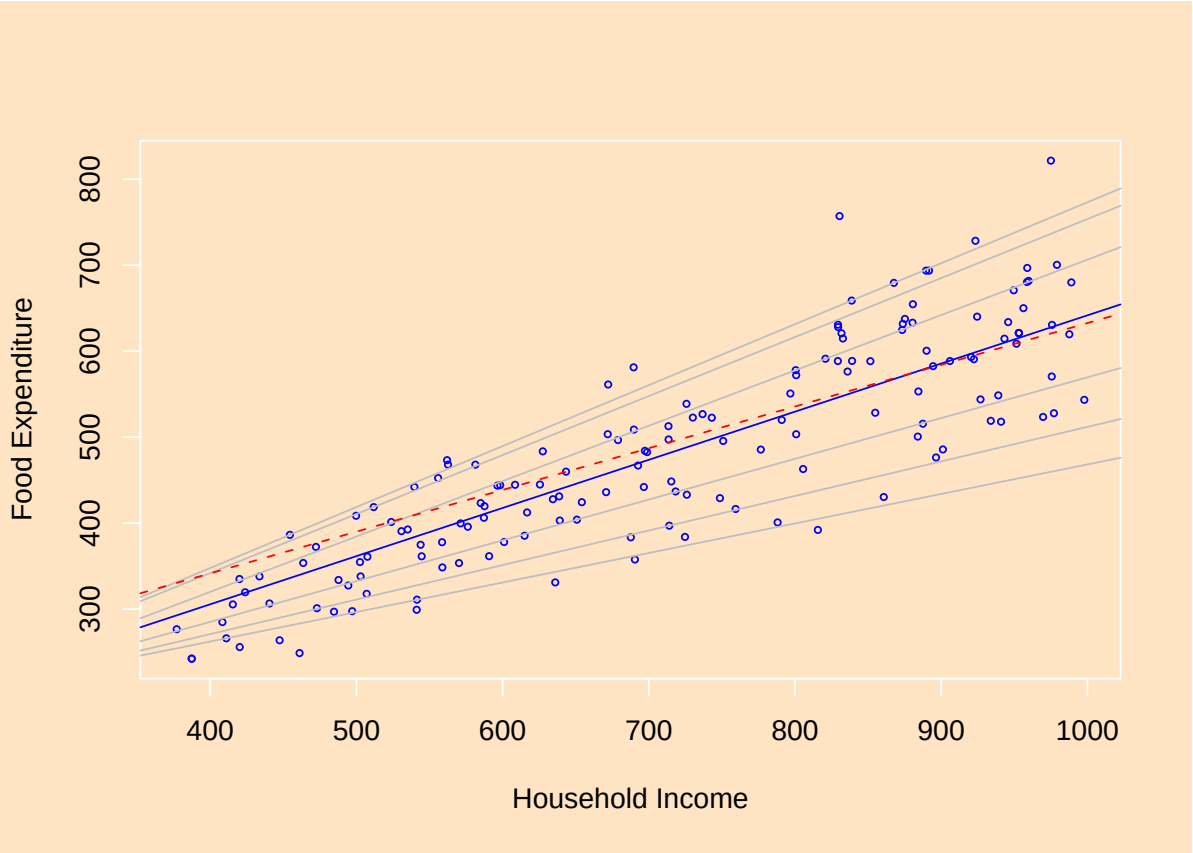
- Engel's Law: A Classical Economic Example
- Infant Birthweight: A Public Health Example
- Melbourne Daily Temperature: A Time Series Example

Engel's Food Expenditure Data



Engel Curves for Food: This figure plots data taken from Engel's (1857) study of the dependence of households' food expenditure on household income. Seven estimated quantile regression lines for $\tau \in \{.05, .1, .25, .5, .75, .9, .95\}$ are superimposed on the scatterplot. The median $\tau = .5$ fit is indicated by the darker solid line; the least squares estimate of the conditional mean function is indicated by the dashed line.

Engel's Food Expenditure Data



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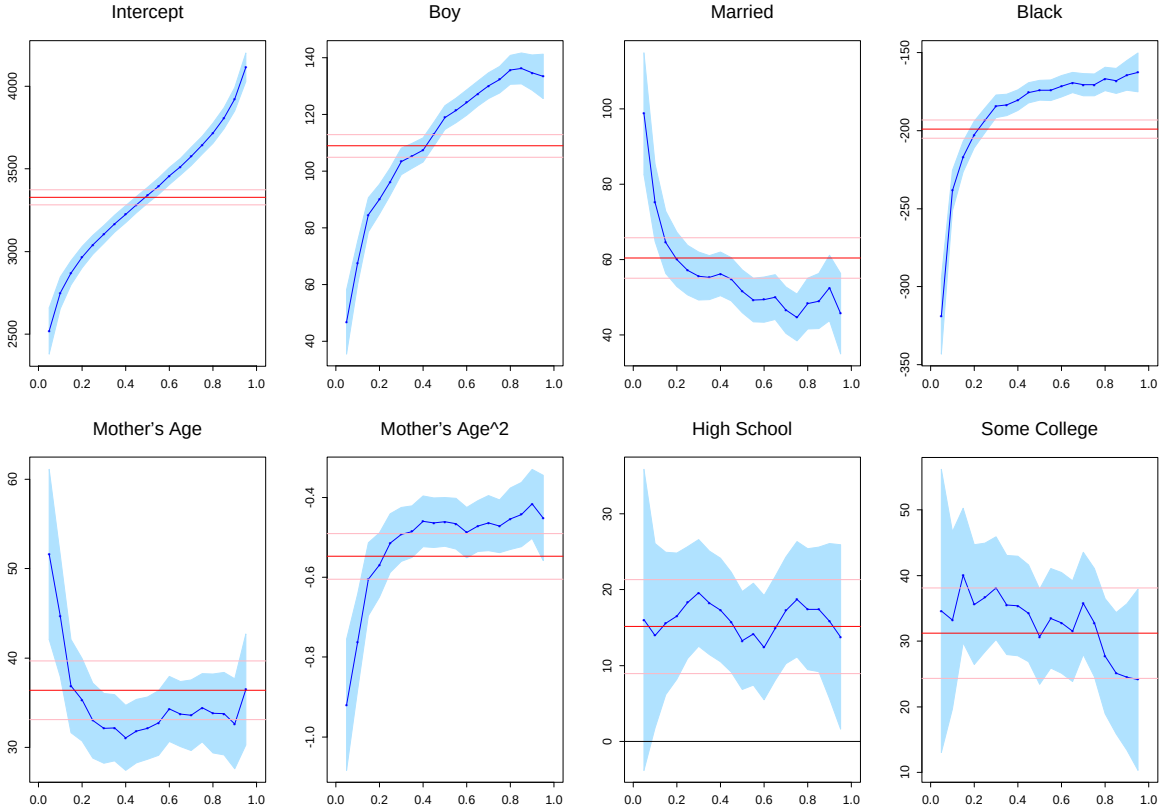
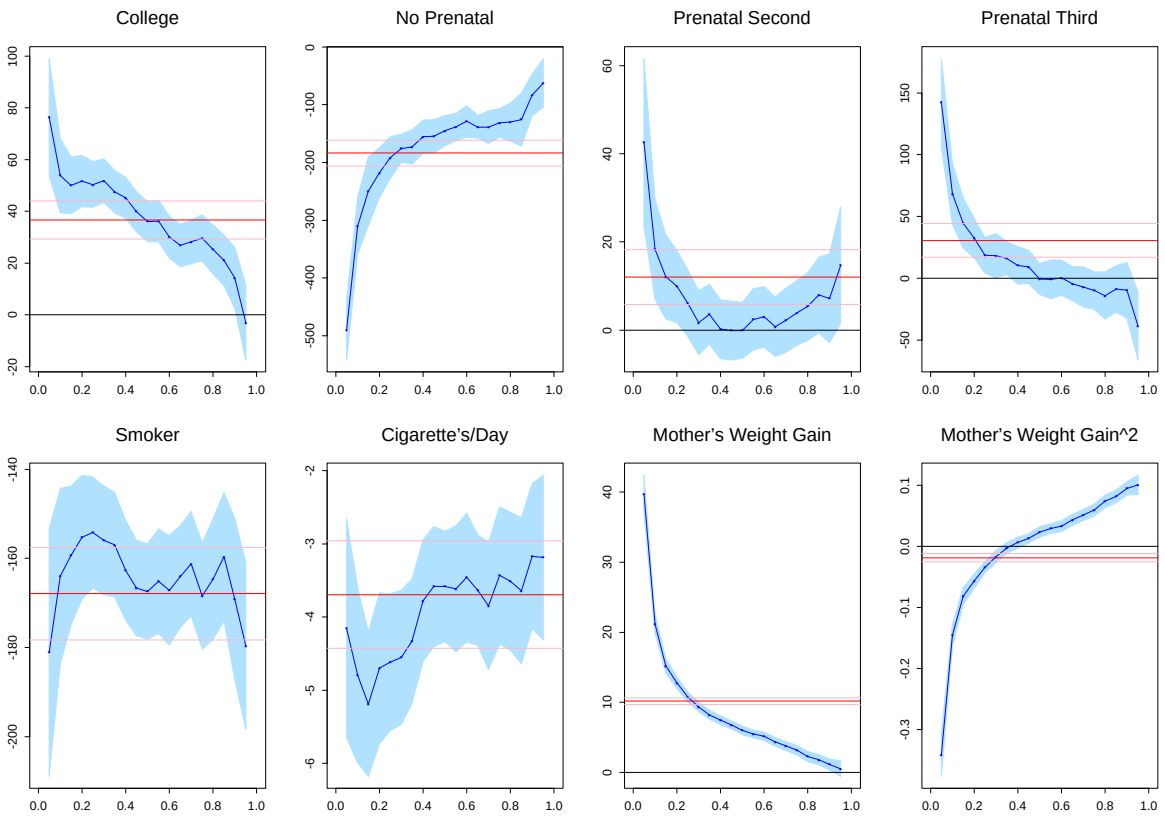
A Model of Infant Birthweight

- Reference: Abrevaya (2001), Koenker and Hallock (2001)
- Data: June, 1997, Detailed Natality Data of the US. Live, singleton births, with mothers recorded as either black or white, between 18-45, and residing in the U.S. Sample size: 198,377.
- Response: Infant Birthweight (in grams)
- Covariates:
  - Mother's Education
  - Mother's Prenatal Care
  - Mother's Smoking
  - Mother's Age
  - Mother's Weight Gain

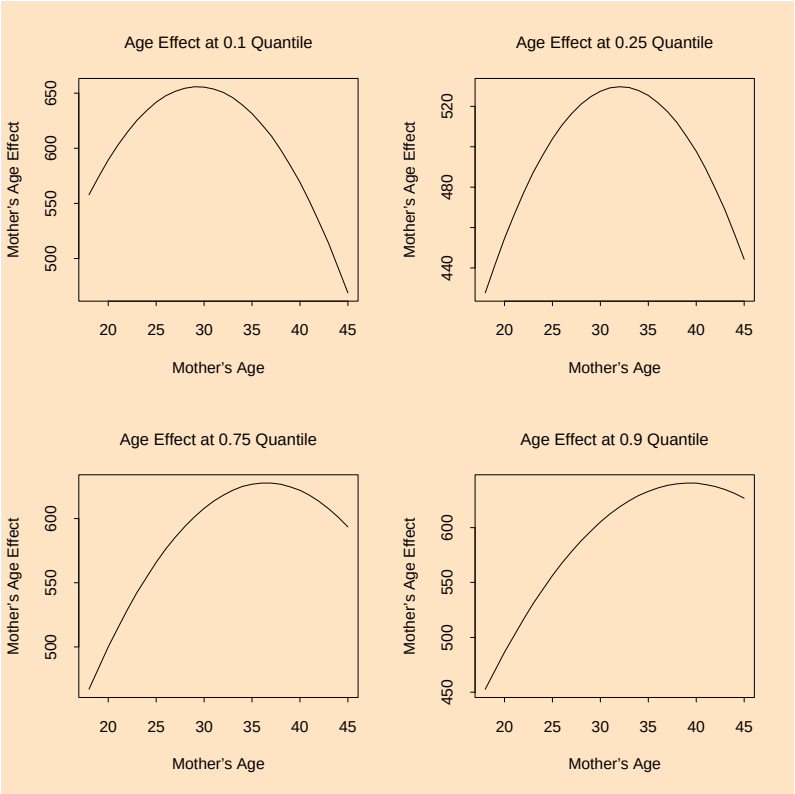
Quantile Regression Birthweight Model I



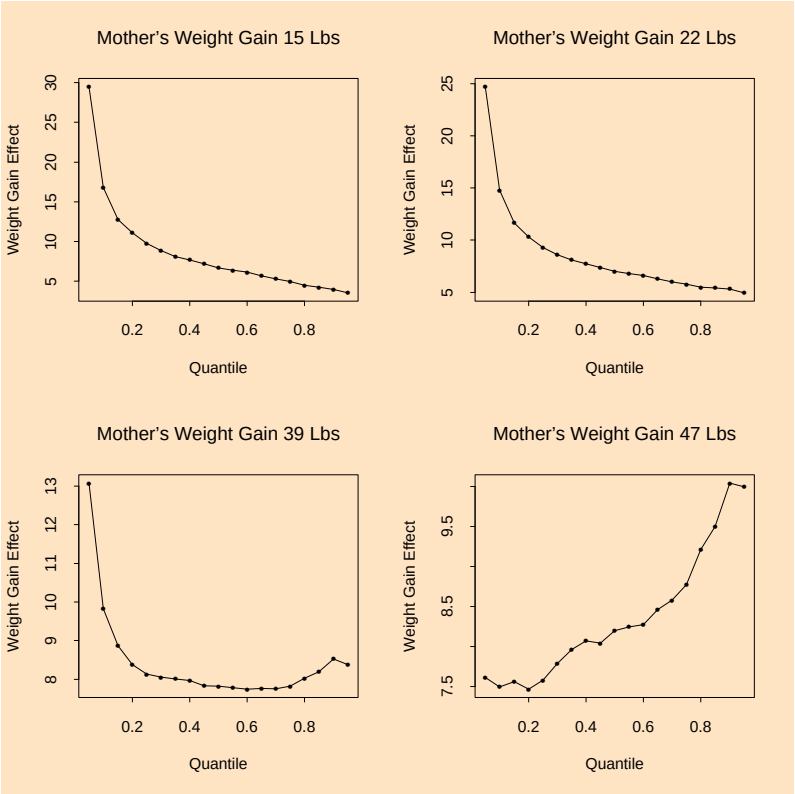
Quantile Regression Birthweight Model II



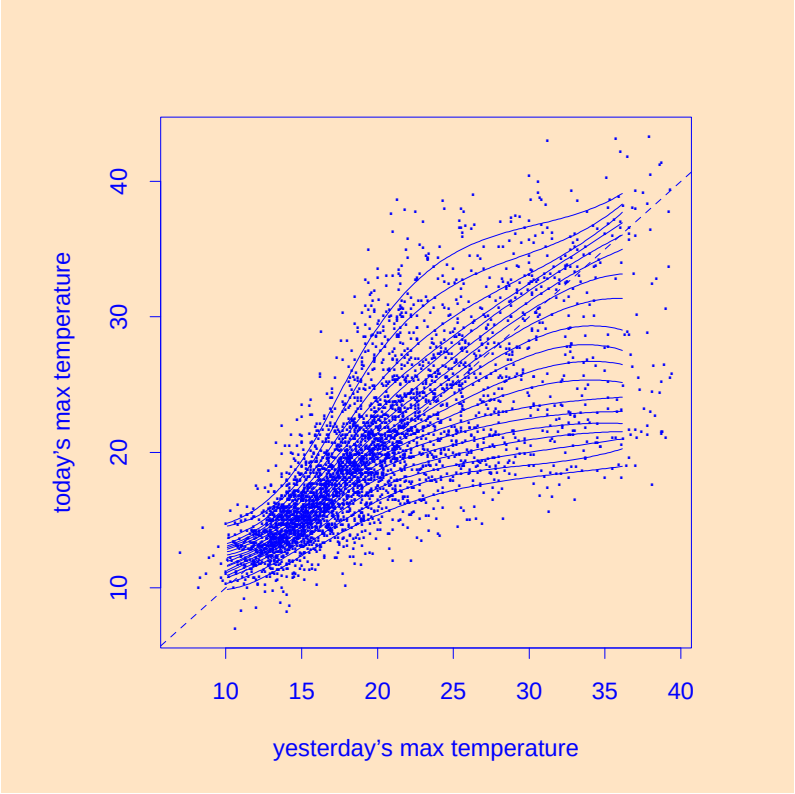
Marginal Effect of Mother's Age



Marginal Effect of Mother's Weight Gain

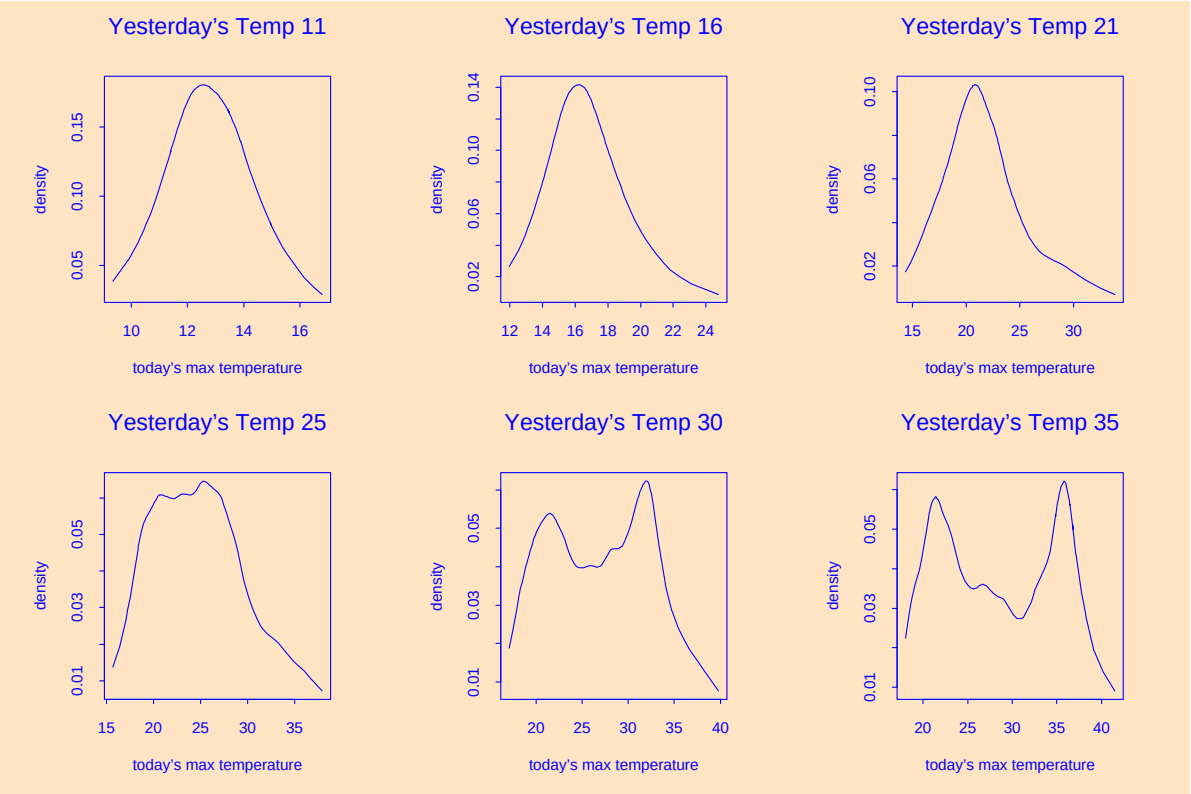


AR(1) Model of Melbourne Daily Temperature



The plot illustrates 10 years of daily maximum temperature data for Melbourne, Australia as an AR(1) scatterplot. Superimposed are estimated conditional quantile functions for $\tau \in \{.05, .10, ..., .95\}$.

Conditional Densities of Melbourne Daily Temperature



Conclusions

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- In applications a variety of unobserved heterogeneity phenomena are rendered observable.